Transmission of Negative Interest Rates: Reversal or Amplification?

Jan Lukas Schäfer[1]

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Abstract

Previous studies show that banks avoid passing negative monetary policy rates through to depositors, causing equity erosion that can constrain lending. This paper identifies a countervailing risk-appetite channel: when the deposit zero lower bound binds, negative rates increase banks' failure risk, strengthening limited-liability incentives to expand risky lending. Consequently, rate cuts below zero can stimulate loan supply more than cuts in positive territory, provided enough banks are unconstrained. A dynamic model calibrated to Germany, the largest euro area economy, finds this effect substantial, increasing aggregate loan supply by about 9% despite equity erosion pressures.

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[1]Banco de España.

E-mail for correspondence: janl.schaefer@bde.es

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1 Introduction

Negative monetary policy rates, long thought to be impossible in practice, have become a reality in many advanced economies since the mid 2010s. The ECB's monetary policy rate, for example, was zero or negative for a decade, with a trough of -0.5%. Commercial banks play a central role in the transmission of monetary policy, and it is of prime interest for central banks to thoroughly study the impact of negative interest rate policy (NIRP) on banks' lending.

A recurrent finding of empirical studies is banks' hesitance to set negative deposit rates, which suggests the existence of a deposit zero lower bound, henceforth *D-ZLB* (see e.g. Heider et al., 2019; Hong and Kandrac, 2021; Basten and Mariathasan, 2023 and Eggertsson et al., 2024, among others). With such a bound, conventional wisdom suggests that negative interest rate policy can be less effective in stimulating bank lending than conventional monetary policy (Ulate, 2021) or even reduce it (Abadi et al., 2023). As banks receive negative rates on their reserves from the central bank, while their ability to pass these costs on to depositors through negative remuneration or higher fees is limited, they make losses in their deposit-taking. This in turn erodes their equity and eventually forces them to reduce lending to comply with regulatory capital requirements.

However, empirical evidence regarding the impact of the D-ZLB on lending is mixed: Some studies find that banks more exposed to negative policy rates increase loan supply vis-a-vis less exposed banks (Hong and Kandrac, 2021; Demiralp et al., 2021; Bottero et al., 2022; Schelling and Towbin, 2022; Grandi and Guille, 2023), others find the opposite result (Heider et al., 2019; Basten and Mariathasan, 2023; Eggertsson et al., 2024; Kwan et al., 2025).[1] But then, how do negative interest rates affect lending by commercial banks? This paper develops a quantitative model in which two different forces interact in opposite directions, making the resulting dominant force contingent on parameters and, thus, providing a possible explanation to the divergent findings in the empirical literature.

[1] Measures of exposure to negative interest rates differ in the empirical literature. For instance, Hong and Kandrac (2021) use the cross-sectional variation in bank's share prices around the central bank's announcement of NIRP. Bottero et al. (2022) use the ex-ante interbank position and liquid assets, while Heider et al. (2019), Eggertsson et al. (2024) and Grandi and Guille (2023) use the amount of deposits relative to total assets. Demiralp et al. (2021) use both central bank reserves in excess of regulatory requirements and retail deposits relative to total assets. In turn, Basten and Mariathasan (2023) exploit tiered remuneration of reserves to construct the share of a bank's reserves affected by NIRP. Finally, Schelling and Towbin (2022) use both the measures of Heider et al. (2019) and Basten and Mariathasan (2023), unlike Eggertsson et al. (2024), who conduct an event study with daily data for loan rates as their main exercise (but cannot do so for loan volumes).

Specifically, this paper identifies and quantifies the importance of a novel amplification channel, labeled *risk-appetite* channel, that affects banks under negative interest rates when (a) they face a D-ZLB, (b) their probability of failure is not zero, and (c) they are not equity-constrained. Through this channel, banks' supply of risky loans increases more strongly after a policy rate cut when their D-ZLB is binding. This mechanism breaks down when banks' capacity to lend is constrained by their equity, in which case the equity-erosion channel emphasized by previous theoretical contributions (Ulate, 2021; Abadi et al., 2023) entirely determines their lending, making it fall as policy rates further decline into negative territory. When the channel identified in this paper dominates, policy rate cuts into negative (or more negative) interest rates would stimulate the economy even more than in positive territory. The dominance of this effect requires that sufficiently many banks are not capital constrained, meaning that they have (or are able to raise) the equity needed to meet capital requirements when they wish to increase their lending.

The risk-appetite channel is related to limited-liability risk-taking distortions that are first established in a stylized static model (henceforth referred to as illustrating model), in which banks are financed by insured deposits, invest in risky loans, and also invest in a safe asset remunerated at the policy rate. Since the bank is exclusively funded with deposits in the illustrating model, that is, there are no capital requirements, the operation of the equity-erosion channel emphasized by the previous literature is shut down in this illustrating model – equity and capital requirements will be introduced in the quantitative model discussed below.

Banks are local monopolists in loan markets, monopolistic competitors in deposit markets (as in e.g. Abadi et al., 2023), and are subject to a lower bound on deposit rates. An exogenously distributed fraction of the risky loans defaults, so that a bank fails whenever the fraction of defaulting loans is large enough to make its residual net worth negative.

In this framework, I show analytically that policy rate cuts are expansionary, unless the representative bank fails with certainty.[2] Further, when the bank's failure probability is also non-zero and the D-ZLB is binding, the D-ZLB induces the bank to charge a lower loan rate for a given policy rate. Under the same conditions, and everything else equal,

[2]The bank may fail with certainty if losses in deposit taking become so high that profits from lending can never exceed them, such that the bank is indifferent between any loan rate. A similar point was raised by Repullo (2020b).

policy rate changes have larger effects on loan rates. The risk-appetite channel is therefore only operative when banks have a non-trivial probability of failure.

Throughout the paper, the deposit rate should be interpreted as the net return on deposits inclusive of fees. Importantly, the risk-appetite channel does not rely on deposit rates being literally bounded at zero. It only requires a decreased pass-through of policy rates to depositors below a certain policy rate. Fees can therefore be interpreted as shifting the effective lower bound on deposit remuneration downward rather than eliminating it altogether. Empirical evidence nevertheless suggests that banks' ability to offset negative policy rates through higher fees has been limited (Heider et al., 2019; Michaelis, 2022; Balloch and Koby, 2023), implying that a lower bound on deposit remuneration remains quantitatively relevant.

The key difference to previous models studying bank lending under negative interest rates – in which the amplification effect through the risk-appetite channel emphasized here did not arise – is that loans are risky, and the bank itself fails when a sufficiently high proportion of its loan portfolio defaults. The amplification effect arises because the losses in deposit taking due to the D-ZLB lead the representative bank to fail for loan default rates for which it otherwise would not have failed. Since the bank is protected by limited liability, its payoff is zero when it fails. Hence, by making the bank fail for lower loan default rates, the D-ZLB shifts losses from the bank to the deposit insurance agency, effectively increasing the conditional-on-bank-solvency profitability of loans for the bank. Thus, the bank charges a lower loan rate when the D-ZLB is binding than in the alternative scenario when it is not.[3] The transmission of monetary policy changes are thus amplified when the D-ZLB is binding, due to the higher (lower) losses in deposit taking when the policy rate is lowered (raised).[4]

The empirical literature commonly explains findings of increased loan supply of banks

[3]Bittner et al. (2022) and Döttling (2025) provide models of bank's monitoring incentives under negative interest rates in which loan defaults are perfectly correlated. The channel emphasized here cannot arise when loan defaults are perfectly correlated, since then losses in deposit-taking do not impact the conditional-on-solvency return on lending: either all loans default and the bank fails or all loans repay and the bank is solvent (for a sufficiently high loan rate) – hence for a given loan rate the D-ZLB either makes the bank fail with certainty (an extreme form of reversal, as noted in Repullo 2020b) or does not impact the bank's failure probability (and thus conditional-on-solvency return on lending) at all.

[4]When the D-ZLB is not binding, the deposit rate R_D is a constant fraction of the policy rate R, such that the deposit spread $R - R_D$ (and thus the bank's profit in deposit taking) is increasing with R. However, the change in the deposit spread is systematically larger once the D-ZLB binds, since then the transmission into deposit rates of changes in R breaks down.

more affected by the D-ZLB as reflecting a search-for-yield or portfolio-rebalancing motive (Hong and Kandrac, 2021; Demiralp et al., 2021; Bottero et al., 2022). The risk-appetite channel identified in this paper provides a microfoundation for such behavior. In the model, the shift toward riskier lending does not arise from a mechanical substitution away from assets with negative yields. Rather, it is an endogenous response to the increase in bank failure risk generated by the interaction of negative policy rates with the D-ZLB. By raising the marginal value of limited liability, this increase in failure risk strengthens banks' incentives to expand risky lending.

After establishing the novel risk-appetite channel, I assess its quantitative relevance relative to the equity-erosion channel emphasized by, among others, Ulate (2021) and Abadi et al. (2023). The equity-erosion channel requires both capital requirements and equity dynamics to be salient: losses in deposit taking erode equity over time until the scarcity of banks' equity forces them to reduce lending to comply with capital requirements. Hence, I develop a dynamic, quantitative extension of the illustrating model in which banks are financed by equity, as well as insured deposits, in order to comply with a regulatory capital requirement (which requires the bank to have a minimum amount of equity funding per unit of lending). The asset side is kept identical to the illustrating model: the bank invests in risky loans and a safe asset remunerated at the monetary policy rate. The latter is exogenous and evolves according to a Markov chain.

As before, in the quantitative model banks have market power in loan and deposit markets, and face a D-ZLB. Loan default risk is modeled following the Vasicek (2002) single risk factor model. Banks are infinitely lived, but are shut down forever if their net-worth becomes negative. This may happen either because of loan defaults, losses in deposit taking or a combination of both.

I assume that banks can raise additional equity at a convex cost, while paying out dividends is costless. Equity dynamics are therefore fully endogenous here, whereas Abadi et al. (2023) assume a constant dividend payout ratio and exogenous equity injections. Thus, in contrast to the illustrating model, in the quantitative model banks may restrict lending when their equity falls low because it is too expensive to raise sufficient equity to support the loan supply level that they would choose if having plenty of available equity – this gives rise to the equity-erosion channel. Apart from their loan pricing decisions, banks are hence able to react to losses from deposits under negative interest rates by

adjusting capitalization through dividend policy and equity injections. These decisions impact on loan supply and the probability of bank failure beyond the more mechanical equity erosion effect of the losses in deposit taking explored in previous studies (e.g. Abadi et al. 2023).

Monetary policy follows a two-state Markov chain, where the policy rate is negative in one state and positive in the other state.

The model is calibrated to Germany, the largest economy in the euro area and home to a large banking sector with significant global systemic relevance, making it a particularly important setting for quantifying the effects of the D-ZLB on bank lending and financial stability.[5][6] The quantitative analysis compares two scenarios: one in which banks are subject to a D-ZLB as observed in the real world (e.g. Heider et al., 2019) and one in which banks can raise deposits at negative deposit rates. Through the lens of the model, the only difference between NIRP and conventional monetary policy is the breakdown of the transmission into deposit rates under NIRP. Hence, the comparison of the two scenarios (D-ZLB vis-a-vis no D-ZLB) serves to assess how the effects of negative interest rate policy differ from conventional monetary policy.

The main conclusions of the quantitative analysis are five. First, banks that are sufficiently capitalized, such that their capital requirement is slack and they would thus not supply higher loan volumes even if they had more capital, supply significantly higher loan volumes under NIRP when subject to a D-ZLB: the loan rate they charge decreases by 11 basis points due to the D-ZLB. Second, 25% of banks supply lower loan volumes due to the D-ZLB during the average spell of negative interest rates, as they are capital-constrained and have to restrict lending due to the equity-eroding effects of losses in deposit taking. However, the first effect dominates in the quantitative results. I find that the aggregate loan supply is on average 9% higher due to the D-ZLB when policy rates are negative. Third, the bank-failure probability is 30 basis points higher under NIRP due to the D-ZLB, up from a baseline level of 47 bps. Fourth, the impact of the D-ZLB on loan

[5]The German banking sector is the second-largest in Europe by assets and the Deutsche Bank is a globally systemically important bank (G-SIB) (European Banking Federation, 2025; Financial Stability Board, 2025).

[6]In Germany, banks have a particularly high deposit-to-loan ratio, whereas in the euro area the average deposit-to-loan ratio is below 1 — in the baseline model this would imply that risky banks have access to non-deposit financing at the riskless interest rate. Since this is inconsistent, and to keep the model parsimonious, the baseline model is calibrated to Germany. As a robustness check, I build an extension of the model in which debt is fairly priced. This extension is calibrated to the euro area. The results are similar.

supply under negative interest rates is hump-shaped over time after a cut from positive to negative rates: the amplification through the risk-appetite channel is strongest initially, and is weakened by the equity-erosion channel over time. Whereas for a cut to -0.31% the risk-appetite channel dominates as long as policy rates remain negative, for a cut to -0.5% the equity-erosion channel starts to dominate after 6 years and bank lending falls below the level that would have prevailed in the absence of a D-ZLB. Nevertheless, both policy rate cuts lead to higher aggregate loan supply even after 50 years. Fifth, these dynamics are more nuanced for cuts within negative territory in the short run and depend on the initial policy rate and the timing of the cuts, while the long run effects of cuts within negative territory are the same as those from positive to negative territory.

Taken together, the results suggest that moderately negative interest rate policy is even more effective than conventional monetary policy at increasing credit supply on average. However, due to capital constraints being binding for some banks, its effects on the cross-section of banks are heterogeneous and the dynamics depend on the initial policy rate and the timing of cuts. Furthermore, NIRP has a sizable cost in terms of financial stability, due to the large increase in the bank-failure probability.

Related Literature This paper contributes to the literature on negative nominal interest rates, reviewed in Balloch et al. (2022), in several ways. First, it contributes to understanding the pass-through of (negative) monetary policy rates into banks' lending rates.[7] The theoretical contributions of Ulate (2021) and Abadi et al. (2023) highlighted that the transmission of negative policy rates into loan rates is impaired due to losses in deposit taking under negative interest rates eroding equity in the presence of a regulatory capital constraint. However, the literature has also emphasized that this equity-erosion channel does not necessarily imply a reduction in bank lending. This is because profits from lending (Ulate, 2021) or reduced funding costs from alternative sources, including bank bonds (Onofri et al., 2023; Eggertsson et al., 2024), can prevent a deterioration in bank equity over time. In models in which this is the case, e.g. in Ulate (2021), the models predict that negative interest rate policy is less effective in stimulating bank lending than

[7]While this paper focuses on the effects of NIRP on bank lending, the literature has identified additional channels that can contribute to expansionary effects of NIRP. These include signaling of prolonged low rates (De Groot and Haas, 2023) and lower yields on corporate and government bonds (Bräuning and Wu, 2017), which maintain the intertemporal substitution channel in households' consumption-saving decision active despite the D-ZLB (Onofri et al., 2023) and lead to lower exchange rates as domestic assets become more attractive for investors vis-a-vis foreign assets (Bräuning and Wu, 2017).

conventional monetary policy. Neither of these theoretical contributions considers the interplay of monetary policy and bank risk. This paper adds to the literature by showing that in this context a novel *risk-appetite* channel arises. Through this channel, risky banks with deposit rates stuck at zero increase their (unconstrained) loan supply more than they would have in absence of the D-ZLB after a policy rate cut. The paper also quantifies the importance of this channel vis-a-vis the equity-erosion channel emphasized by previous contributions.

As discussed above, the empirical literature has found mixed evidence regarding the pass-through of NIRP into lending rates. This paper contributes to understanding these divergent findings. Concretely, the amplification of monetary policy transmission through the risk-appetite channel when the D-ZLB is binding rationalizes empirical findings of increased loan supply of banks more exposed to negative interest rates (Hong and Kandrac, 2021; Demiralp et al., 2021; Bottero et al., 2022; Schelling and Towbin, 2022; Grandi and Guille, 2023). Through the lens of previous models of negative interest rates that abstracted from risky lending (Ulate, 2021; Darracq Paries et al., 2023; Onofri et al., 2023; Eggertsson et al., 2024; Abadi et al., 2023), increased lending of banks more exposed to NIRP as documented in different contexts is difficult to rationalize. While in Abadi et al. (2023), Onofri et al. (2023) and Eggertsson et al. (2024) monetary policy cuts can increase bank equity, at least in the short run, these effects do not operate differently under negative interest rate policy than under conventional monetary policy.[8] Hence, such explanations would arguably be at odds with placebo-exercises confirming abnormal effects of NIPR (Hong and Kandrac, 2021; Heider et al., 2019).

This paper also contributes to the literature on the financial stability implications of negative interest rate policy. The empirical literature has consistently found evidence for an increase in riskiness of more exposed banks (Nucera et al., 2017; Hong and Kandrac, 2021; Basten and Mariathasan, 2023; Heider et al., 2019).[9] Further, evidence for portfolio reallocation towards riskier assets is found by Bottero et al. (2022); Basten and Mariathasan (2023); Schelling and Towbin (2022); Heider et al. (2019). However, the

[8] In Abadi et al. (2023) such an increase in bank equity can arise due to capital gains in fixed-rate long-term assets. In Onofri et al. (2023) and Eggertsson et al. (2024), it can arise because of a reduction in the cost of non-deposit funding.

[9] While Nucera et al. (2017) do not provide a differences-in-differences analysis, they show an increase in their measure of bank risk for banks with business-models associated with a higher deposit dependence, such as retail banks and mutual/co-operative type banks. On the other hand, they document a decrease in the measure of riskiness for large universal banks.

financial stability implications of negative interest rates have not been studied through the lens of a quantitative model. The risk-appetite channel emphasized here arises in the model because of an increase in the bank's failure probability, and implies a reallocation of assets from the safe asset to risky loans. Hence, the model provides a framework that jointly explains the empirical findings of an increased bank-failure probability, a portfolio-reallocation towards riskier assets and an increase in bank lending.[10]

While the focus of this paper is highlighting the novel risk-appetite channel, the model is not inconsistent with contractionary effects of negative interest rate policy (Heider et al., 2019; Basten and Mariathasan, 2023; Eggertsson et al., 2024; Kwan et al., 2025). Through the lens of the model, such effects could reflect (a) portfolio reallocation away from the particular type of loans studied in a given empirical contribution (e.g. away from syndicated loans in Heider et al., 2019), or (b) a sufficiently large fraction of banks constrained by their capital requirement.[11]

Outline The rest of the paper is organized as follows: Section 2 presents a stylized model to study the loan supply of banks unconstrained by capital requirements under negative interest rates. Section 3 presents the quantitative model. The model's calibration is described in Section 3.2, while Section 3.3 tests an empirical prediction. Section 3.4 presents the results of the numerical analysis. Section 4 discusses the results of both the stylized and the quantitative model, and relates them to the empirical literature, while Section 5 concludes. The Online Appendix contains additional tables and figures, as well as proofs of all lemmas, propositions and corollaries.

[10]Regarding the increased bank-failure probability, the mechanism exposed here is complementary to existing explanations of increased risk taking linked to limited liability (Repullo, 2004; Dell'Ariccia et al., 2014).

[11]To be more precise, the diverging findings for the EA of Heider et al. (2019) (lower loan supply *in the syndicated loan market* of more exposed banks) and Demiralp et al. (2021) (higher loan supply *overall* of more exposed banks) might point to a portfolio reallocation away from syndicated loans, rather than effects due to capital requirements. Eggertsson et al. (2024) focus on lending to households and Kwan et al. (2025) on mortgages, hence it cannot be ruled out that their result of a decrease in these loan products reflects a portfolio reallocation, rather than a decrease in total lending. Schelling and Towbin (2022) argue that the diverging findings for Switzerland in their paper and Basten and Mariathasan (2023) possibly reflect heterogeneous demand effects, which cannot be controlled for in the latter study.

2 The Amplification Mechanism in a Simple Model

This section illustrates in the context of a simple static partial equilibrium model how the presence of a D-ZLB can amplify (rather then dampen) the impact of interest rate cuts on bank lending. In this model, the presence of the D-ZLB implies that banks cannot fix gross deposit rates below some $\underline{R}_D$ (so $\underline{R}_D = 1$ corresponds to to the typical case in which net deposit rates cannot be negative). While Section 2.1 establishes the novel channels that operate in the quantitative model of Section 3 formally, Section 2.2 discusses and illustrates some additional results regarding monetary policy transmission with a numerical example.

2.1 Static Model

Consider an individual bank that operates between two dates $t = 0, 1$. At $t = 0$ the bank is financed by insured deposits D remunerated at a gross rate R_D, invests in risky loans L that pay a gross loan rate R_L and may also invest in a safe asset S remunerated at the policy rate R.

The bank is a monopolist in a local loan market where it faces a downward sloping iso-elastic loan demand function of the form $L(R_L) = AR_L^{-\epsilon_L}$, $\epsilon_L > 1$. At $t = 1$, a share ω of its loans default, in which case the corresponding payoff is zero. The random variable ω is assumed to have full support in $[0, 1]$ and its pdf and cdf will be denoted by $f(\omega)$ and $F(\omega)$, respectively. To isolate the implications of the mechanism that I aim to illustrate, the bank faces no capital requirement.

As in Abadi et al. (2023), I assume that the bank competes in deposit markets in a monopolistic fashion with a unit continuum of identical other banks. Thus, the bank is facing an upward sloping deposit function $D(R_D) = \left(\frac{R_D}{\mathcal{R}_D}\right)^{-\epsilon_D} \bar{D}$, with $\epsilon_D < -1$ and $\bar{D} > 0$, where $\mathcal{R}_D$ is the CES index of deposit rates, taken as given by each individual bank. Banks cannot offer deposit rates below $\underline{R}_D$, reflecting the D-ZLB.

Each bank is managed in the interest of its risk neutral shareholders, who enjoy limited liability. When a bank defaults (henceforth: fails) at $t = 1$, its assets are repossessed by a deposit insurance agency, who pays all its depositors in full. The representative bank

maximizes its expected value at $t = 1$:

$$\max_{R_L, R_D \geq \underline{R}_D, S \geq 0} \int_0^1 \max\{(1-\omega) R_L L(R_L) + RS - R_D D(R_D), 0\} dF(\omega), \tag{1}$$

subject to the balance sheet constraint:

$$L(R_L) + S = D(R_D). \tag{2}$$

To solve the bank's optimization problem, I will proceed in two steps. The constraint $S \geq 0$ is guessed to be non-binding, which can ex-post be shown to imply a parameter restriction on $\bar{D}$. Using this guess to substitute S from (2.2) into the objective function (2.1), the objective function can be written as:

$$\max_{R_L, R_D \geq \underline{R}_D} \int_0^1 \max\{(R_L - R) L(R_L) + (R - R_D) D(R_D) - \omega R_L L(R_L), 0\} dF(\omega). \tag{3}$$

First the optimal deposit rate can be characterized as independent of the bank's choice of R_L: let it be denoted $\hat{R}_D$. The results regarding the optimal deposit rate are summarized in the following lemma:

Lemma 1. *Assuming the bank does not always fail, the bank's optimal deposit rate is:*

$$\hat{R}_D(R) = \max[R_D^*(R), \underline{R}_D], \tag{4}$$

where

$$R_D^*(R) = \frac{\epsilon_D}{\epsilon_D - 1} R \tag{5}$$

is the bank's optimal deposit rate in the absence of a D-ZLB. Moreover, $\hat{R}_D(R) = R_D^*(R)$ for all $R \geq R^*$ and $\hat{R}_D(R) = \underline{R}_D$ for all $R < R^*$, where $R^* = (\epsilon_D - 1)/\epsilon_D \underline{R}_D$.

By symmetry, all monopolistic competitors in the deposit market offer the same deposit rate. Hence the CES index of deposit rates is $\mathcal{R}_D = \hat{R}_D(R)$ and an individual bank's profits from deposits are given by $[R - \hat{R}_D(R)]\bar{D}$.

Next, it will be shown that the deposit rate does impact the optimal loan rate, and therefore monetary policy transmission, if the D-ZLB is binding and the bank has an

interior failure probability. The channel works through the impact that the deposit rate has on such a probability. The bank fails if its residual net-worth is negative:

$$(R_L - R)L(R_L) + (R - R_D)\bar{D} - \omega R_L L(R_L) < 0,$$

that is if:

$$\omega > \tilde{\omega} = \frac{(R_L - R)L(R_L) + (R - \hat{R}_D)\bar{D}}{R_L L(R_L)}, \tag{6}$$

and is otherwise solvent. When $\tilde{\omega} > 1$ the bank never fails since ω has support in the interval $[0, 1]$ – this could happen if profits from deposits $(R - \hat{R}_D)\bar{D}$ are so large that the bank can repay $RL(R_L)$ even if all loans default. Of course, $\tilde{\omega} > 1$ cannot happen under policy rates $R < \underline{R}_D$; however, when $R < \underline{R}_D$ profits from deposits may be sufficiently negative, such that $\tilde{\omega} < 0$, and the bank always fails.

Using previous results, it is convenient to write the objective function of the bank as:

$$\Pi(R_L) = \int_0^{\bar{\omega}} \{[R_L(1 - \omega) - R]L(R_L) + (R - \hat{R}_D)D(\hat{R}_D)\}dF(\omega), \tag{7}$$

where $\bar{\omega} = \min[\max(\tilde{\omega}, 0), 1]$, that is:

$$\bar{\omega} = \begin{cases} 0 \text{ if } \tilde{\omega} < 0 \\ \tilde{\omega} \text{ if } 0 \leq \tilde{\omega} \leq 1 \\ 1 \text{ if } \tilde{\omega} > 1. \end{cases} \tag{8}$$

Then, the following proposition characterizes the optimal loan rate:

Proposition 1. *If there exists a R_L such that $(R_L - R)L(R_L) + (R - \hat{R}_D)D(\hat{R}_D) > 0$, the problem $\max_{R_L} \Pi(R_L)$ has a global maximum at $\hat{R}_L \in (R, \infty)$, which fulfills:*

$$(1 - \omega^{LL})\hat{R}_L = \frac{\epsilon_L}{\epsilon_L - 1}R, \tag{9}$$

where ω^{LL} is the expected loan default rate conditional on bank solvency:

$$\omega^{LL} = \mathbb{E}[\omega \mid \omega \leq \bar{\omega}] = \frac{1}{F(\bar{\omega})}\int_0^{\bar{\omega}} \omega dF(\omega). \tag{10}$$

The condition for existence of an optimal loan rate $\hat{R}_L \in (R, \infty)$ is always satisfied

under policy rates above the lower bound on deposits $\underline{R}_D$. However, when the bank makes too high losses in deposit taking, such that profits from lending can never exceed these losses, the bank always fails. In that case, the bank is indifferent between any loan rate.

In general, it is neither guaranteed that the solution to (9) is unique, nor that it involves $\Pi''(\hat{R}_L) \neq 0$ (see Online Appendix D).[12]. In what follows it is assumed that parameters are such that a unique $\hat{R}_L$ solving (9) exists, and $\Pi''(\hat{R}_L) < 0$. This is required to use the implicit function theorem to characterize changes in the optimal loan rate (see Online Appendix A.3).

Using (6) and Lemma 1, it follows that a stricter lower bound $\underline{R}_D$ on the deposit rate decreases the cutoff $\tilde{\omega}$ for any given loan rate when the lower bound is binding:

$$\frac{\partial \tilde{\omega}}{\partial \underline{R}_D}\Big|_{\hat{R}_L} = \begin{cases} -\dfrac{\bar{D}}{\hat{R}_L L(\hat{R}_L)} \text{ if } R < R^* \\ 0 \text{ if } R > R^*, \end{cases} \tag{11}$$

which follows from (6) and Lemma 1. By the definition of $\bar{\omega}$, the D-ZLB thus increases the banks' failure probability for a given loan rate, whenever that failure probability is interior (as opposed to strictly zero or one).[13] This in turn decreases the conditionally expected loan default rate ω^{LL} by (10): a stricter lower bound means that (for a given loan rate) the bank fails for the highest loan default rates for which the bank would otherwise have been solvent. This lowers the loan default rate ω^{LL} that the bank expects conditional on its own solvency.

Whenever the D-ZLB induces such a decrease in ω^{LL}, Equation (9) suggests that it decreases the loan rate charged by the bank: intuitively, due to limited liability, the bank only cares about the return on lending *conditional-on-solvency*, hence the lower payoff from deposit-taking induced by the D-ZLB makes risky lending more attractive.

[12]While there is a unique local maximum $\hat{R}_L$ in all my numerical simulations, the sign of $\Pi''(R_L)$ is generally ambiguous such that multiple solutions of (9) might arise when ω^{LL} is increasing in R_L. A unique local maximum involving $\Pi''(\hat{R}_L) \neq 0$ is for example guaranteed when $\omega \sim Unif[0,1]$ and $R \neq \underline{R}_D$ (see Online Appendix D)

[13]Technically:

$$\frac{\partial \bar{\omega}}{\partial \underline{R}_D}\Big|_{\hat{R}_L} \begin{cases} = 0 \text{ if } R > R^* \text{ or } \tilde{\omega} > 1 \text{ or } \tilde{\omega} < 0 \\ \text{does not exist if } R = R^* \text{ or } \tilde{\omega} = 0 \text{ or } \tilde{\omega} = 1 \\ < 0 \text{ else} \end{cases}$$

The non-existence of the partial derivative if $\tilde{\omega} \in \{0,1\}$ is due to the non-differentiability of the function $max(x,0)$ at 0 and the function $min(x,1)$ at 1.

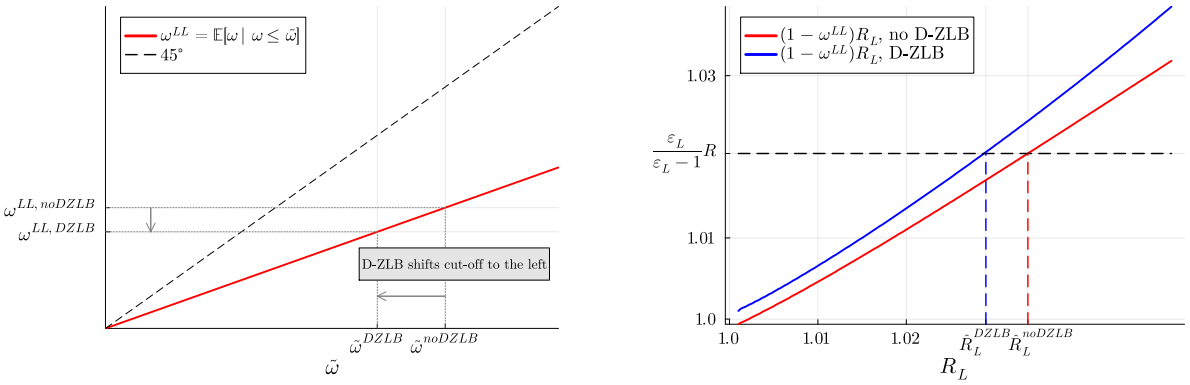


Figure 1: *Illustration of the Risk-Appetite Channel*

This *risk-appetite* channel is thus similar to the mechanism emphasized by Bahaj and Malherbe (2020) in a different context – this parallel is further explored in Section 2.3. The risk-appetite channel is illustrated in Figure 1 and summarized in the following proposition:

Proposition 2. *Assume a unique interior solution in $\hat{R}_L$ exists such that $\Pi''(\hat{R}_L) < 0$ and the bank has a non-trivial probability of failure ($\tilde{\omega} \in (0,1)$). Then a stricter deposit lower bound $\underline{R}_D$ lowers the optimal loan rate if and only if the D-ZLB constraint is binding:*

$$\frac{\partial \hat{R}_L}{\partial \underline{R}_D} \begin{cases} < 0 \text{ if } R < R^* \\ = 0 \text{ if } R > R^*. \end{cases} \tag{12}$$

The proposition implies that when a bank subject to a D-ZLB has an interior failure probability, a cut from any rate above R^* (where the D-ZLB is not binding) to any rate below R^* (where the D-ZLB is binding) is more effective in stimulating lending than in the alternative scenario when banks are not subject to a D-ZLB. To see this, note that the scenario in which banks are not subject to a D-ZLB corresponds to the case $\underline{R}_D \to -\infty$. For policy rates above R^*, banks charge the same loan rate in both scenarios (since the loan rate does not change with the lower bound on deposits when $R > R^*$), while for policy rates below R^* banks will charge a lower loan rate in the scenario in which they are subject to a D-ZLB (since when $R < R^*$ the loan rate decreases when the lower bound on deposits increases).[14]

However, while monetary policy cuts into policy rates below the deposit lower bound

[14] In the real world scenario in which banks do not fix negative deposit rates ($\underline{R}_D = 1$), such a cut from a policy rate above R^* to a rate below R^* may or may not be a cut into negative policy rates, since $R^* = \frac{\epsilon_D - 1}{\epsilon_D} > 1$.

are (ceteris paribus) more expansionary, such cuts have adverse implications for bank failure probabilities. It was already discussed above that $\bar{\omega}$, and thus the bank's failure probability, is decreased by the D-ZLB *for a given loan rate*. Indeed, as a corollary to Proposition 1, it can be shown that the D-ZLB decreases $\bar{\omega}$ unconditionally, when it is interior:

Corollary 1. *Assume a unique interior solution in $\hat{R}_L$ exists such that $\Pi''(\hat{R}_L) < 0$ and the bank has a non-trivial probability of failure ($\tilde{\omega} \in (0,1)$), then the D-ZLB lowers the bank's failure threshold:*

$$\frac{\partial \bar{\omega}}{\partial \underline{R}_D} = \underbrace{\frac{\partial \bar{\omega}}{\partial R_D}}_{<0} \frac{\partial R_D}{\partial \underline{R}_D} + \underbrace{\frac{\partial \bar{\omega}}{\partial R_L}}_{>0} \underbrace{\frac{\partial R_L}{\partial \underline{R}_D}}_{\leq 0} \begin{cases} < 0 \text{ if } R < R^* \\ = 0 \text{ if } R > R^*. \end{cases} \tag{13}$$

This is due to a combination of the direct impact on the bank's failure probability of paying higher deposit rates, and the impact of the lower loan rate the bank offers as implied by Proposition 2.

While the results in Proposition 1 and Corollary 1 are those that operate in the quantitative model of Section 1, some additional results regarding monetary policy transmission are discussed and illustrated in the following subsection.

2.2 Monetary Policy Transmission: A Numerical Example

A numerical example illustrating monetary policy transmission in alternative scenarios is provided in Figure 2, which plots the loan volume $L(\hat{R}_L)$, the loan rate $\hat{R}_L$ and the bank's failure probability (EDF) against the policy rate R. The figure compares the baseline scenario in which banks may not set deposit rates below 1 ($\underline{R}_D = 1$) with a counterfactual scenario in which banks are not subject to a lower bound on deposits, and hence always set $R_D = R_D^*$ (as defined in Lemma 1).

In the left column of the figure, the bank's payoff is deterministic and the bank's failure probability is thus zero (that is $Pr(\omega < \bar{\omega}) = 0$), while in the right column banks are risky (that is $Pr(\omega < \bar{\omega}) \in (0,1)$). To this end, the numerical example assumes that the loan default distribution follows the Vasicek (2002) distribution with

$$F(\omega) = \Phi\left(\frac{\sqrt{1-\rho}\Phi^{-1}(\omega) - \Phi^{-1}(p)}{\sqrt{\rho}}\right), \tag{14}$$

where $\Phi(\cdot)$ is the standard normal cdf, and p and ρ are parameters, governing the borrowers' unconditional default probability and the correlation between their defaults, respectively. I set $p = 0.01$ (i.e., a 1% loan default probability). The cdf $F(\omega)$ converges to a one-point distribution as $\rho \to 0$ (Vasicek, 2002), such that banks have a deterministic payoff in the limit. Thus, for risky banks, I set $\rho = 0.2$, while for safe banks $\rho \to 0$.[15]

The remaining parameters used for this numerical example are $\epsilon_D = -100$ (implying a deposit spread of 1% when the D-ZLB is not binding), $\bar{D} = 2.5$, $\epsilon_L = 50$ (implying a loan rate spread of about 3% for safe banks) and $A = 4.17$ (implying that the constraint $S > 0$ is always satisfied).

As Figure 2 shows, the loan rate is increasing in the policy rate in all scenarios. Hence, there is no reversal. Indeed, it can be shown that:

Corollary 2. *Assume a unique interior solution in $\hat{R}_L$ exists such that $\Pi''(\hat{R}_L) < 0$ and the bank does not fail with certainty, i.e. $(\tilde{\omega} \in (0, 1])$. Then, if*

$$\frac{\partial \omega^{LL}}{\partial \bar{\omega}} < \frac{\epsilon_L}{\epsilon_L - 1}, \tag{15}$$

monetary policy is expansionary:

$$\frac{\partial \hat{R}_L}{\partial R} > 0 \quad \textit{for all } R \neq R^*. \tag{16}$$

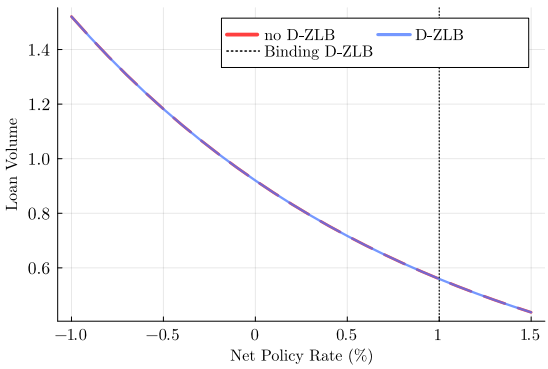
The condition (15) is sufficient but not necessary for monetary policy to be expansionary.[16] The first part of Corollary 2 then implies that monetary policy cuts are always expansionary – and thus there is no reversal – as long as banks do not fail with certainty (in which case the bank is indifferent between any loan rate).

The right panel of Figure 2 highlights a clear kink of monetary policy transmission at $R = R^*$ (the vertical dotted line). There is no such kink in the left panel. The figure thus emphasizes that the D-ZLB only amplifies monetary policy cuts into negative rates when banks have an interior failure probability $Pr(\omega < \bar{\omega}) \in (0, 1)$.

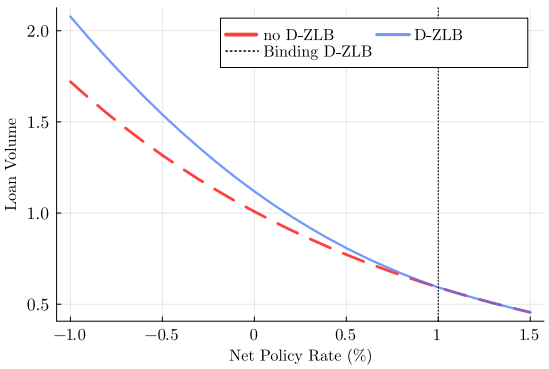
Moreover, the figure shows that when banks have such an interior failure probability, the loan rate in the D-ZLB scenario falls further and further below the loan rate in the

[15] The Basel II regulatory framework (Basel Committee on Banking Supervision, 2004) specifies a formula for the parameter ρ based on the loan default probability p, which would yield $\rho \approx 0.19$ for $p = 0.01$.

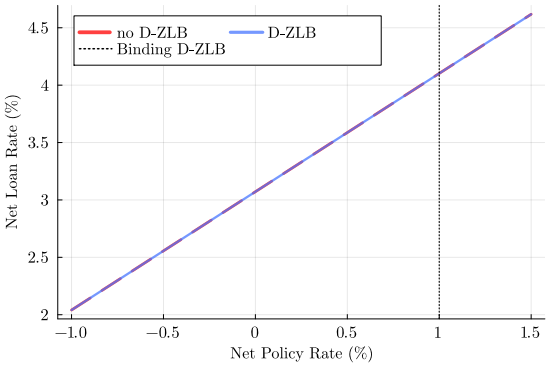
[16] The condition is for example fulfilled for the uniform distribution.



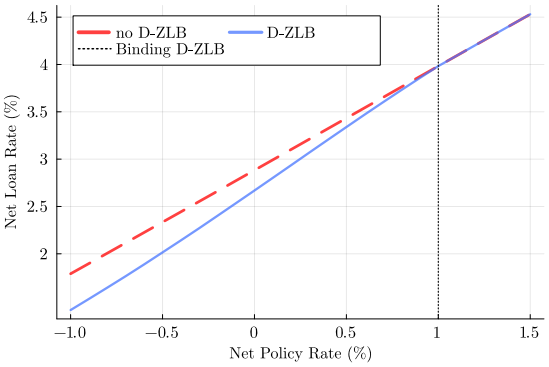
(a) *Loan Volume Safe Banks*



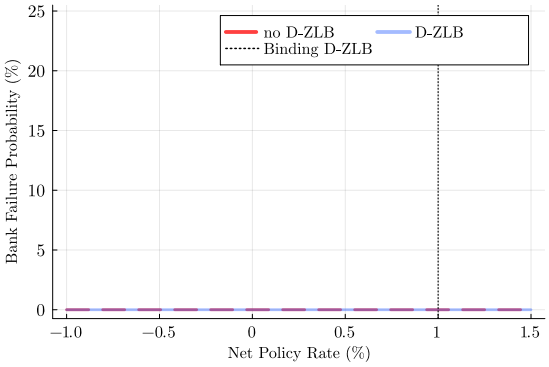
(b) *Loan Volume Risky Banks*



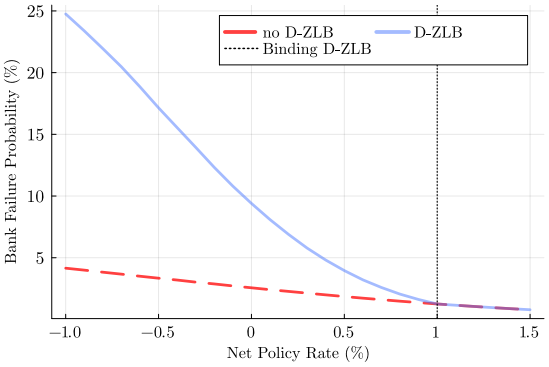
(c) *Loan Rate Safe Banks*



(d) *Loan Rate Risky Banks*



(e) *Expected Failure Probability Safe Banks*



(f) *Expected Failure Probability Risky Banks*

Figure 2: *Numerical Example, Stylized Model*

Note: In the left column (safe banks), the loan default probability is 1% *with probability 1, while in the right column (risky banks) the loan default probability follows a Vasicek (2002) distribution with mean* 1%.

counterfactual scenario without D-ZLB. The reason is a fundamental difference in the transmission of monetary policy rates above and below R^*, as shall be discussed below.[17] The figure shows that the failure probability of risky banks begins increasing once the D-ZLB is binding relative to the failure probability prevailing in the alternative scenario in which the bank sets $R_D = R_D^*$, reflecting Corollary 1. It can be conjectured from (9) that increased bank riskiness (everything else equal) lowers the loan rate but by the same token weakens monetary policy transmission.[18] However, despite the sizable decrease in the bank failure threshold shown in Figure 2, monetary policy transmission is almost linear below R^* – suggesting that the impact of the level of bank riskiness on monetary policy transmission is limited.

To understand the reason for the fundamental difference of transmission of policy rates above and below R^*, it is useful to consider two scenarios in which a bank faces the same policy rate and has the same failure probability before the policy rate cut, and is subject to a binding D-ZLB in one scenario but not in the other. An increase in the policy rate makes banks safer in both scenarios and thus pushes up loan rates further (by Eq. (9)) through changes in the deposit rate spread $R - \hat{R}_D$. By Lemma 1, these are

$$\frac{\partial(R - \hat{R}_D)}{\partial R} = \begin{cases} 1 - \frac{\epsilon_D}{\epsilon_D - 1} \in (0,1) \\ 1 \text{ if } R < R^*. \end{cases}.$$

It is then clear that the change in the deposit spread, and therefore its effect on monetary policy transmission via the change in the bank-failure probability, is particularly large in the scenario in which the D-ZLB is binding, such that $\hat{R}_D = \underline{R}_D$ and hence the deposit spread changes one-to-one with the policy rate. Formally, the following corollary can be proven:

Corollary 3. *Assume a unique interior solution in $\hat{R}_L$ exists such that $\Pi''(\hat{R}_L) < 0$ and the bank has a non-trivial probability of failure ($\tilde{\omega} \in (0,1)$). Consider two different lower bounds $\underline{R}_{D1}$, $\underline{R}_{D2}$ with $\underline{R}_{D1} > \underline{R}_{D2}$ such that the lower bound constraint would be binding under $\underline{R}_{D1}$ (that is $R^*(\underline{R}_{D1}) > R$), but not under $\underline{R}_{D2}$ (that is $R^*(\underline{R}_{D2}) < R$). Everything*

[17] This fundamental difference is also the reason for the non-existence of the derivative $\frac{\partial \hat{R}_L}{\partial R}$ at $R = R^*$ implied by Corollary 2.

[18] The exact impact of a change in the bank-failure probability on monetary policy transmission depends on how the slope of the conditional expectation $\frac{\partial \omega^{LL}}{\partial \bar{\omega}}$ changes with $\bar{\omega}$.

else equal, a policy rate cut stimulates bank lending more when the lower bound constraint is binding:

$$\left.\frac{\partial \hat{R}_L}{\partial R}\right|_{\underline{R}_{D1},R,\bar{\omega}} - \left.\frac{\partial \hat{R}_L}{\partial R}\right|_{\underline{R}_{D2},R,\bar{\omega}} = \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{\bar{D}}{L(R_L)\left[(1-\omega^{LL}) - R_L \frac{\partial \omega^{LL}}{\partial R_L}\right]} \frac{\epsilon_D}{\epsilon_D - 1} > 0. \quad (17)$$

While the conditioning on both the policy rate and the bank failure cutoff in Corollary 3 is counterfactual for policy rates $R \neq R^*$ in the model, it is nevertheless useful to understand the fundamental difference in the transmission of policy rates above and below the zero lower bound, as shown in Figure 2.[19]

2.3 Intuition: D-ZLB Decreases Limit Liability Subsidy

The risk-appetite channel established above can be alternatively interpreted in terms of changes in the marginal limited liability subsidy, similar to the mechanism emphasized by Bahaj and Malherbe (2020) in the context of legacy loans. To gain intuition about this, marginal profits of raising the loan rate are (by the Leibniz rule):

$$\Pi'(R_L) = \int_0^{\bar{\omega}} \{[R_L(1-\omega) - R]L'(R_L) + (1-\omega)L(R_L)\} dF(\omega). \quad (18)$$

The marginal profits can be decomposed into the marginal profits that a bank without limited liability would obtain and a marginal limited liability subsidy, in a similar fashion to Bahaj and Malherbe (2020):

$$\begin{aligned}\Pi'(R_L) &= [R_L(1-\mathbb{E}\omega) - R]L'(R_L) + (1-\mathbb{E}\omega)L(R_L) \\ &\underbrace{- \int_{\bar{\omega}}^{1} \{[(R_L(1-\omega) - R]L'(R_L) + (1-\omega)L(R_L)\} dF(\omega)}_{\equiv MS}. \end{aligned} \quad (19)$$

where the second line of (19), which integrates over the bank's failure region, is the marginal limited liability subsidy MS. It is obvious that the failure cutoff (and hence the

[19] In the context of the model, keeping the failure cutoff fixed can be thought of as banks receiving an exogenous compensation $\Delta = \bar{D}\left(\underline{R}_{D1} - \frac{\epsilon_D}{\epsilon_D - 1}R\right)$ from the government, such that:

$$\bar{\omega} = \frac{(R - \underline{R}_{D1})\bar{D} + (R_L - R)L(R_L) + \Delta}{R_L L(R_L)} = \frac{(1 - \frac{\epsilon_D}{\epsilon_D - 1})R\bar{D} + (R_L - R)L(R_L)}{R_L L(R_L)}$$

and hence $\bar{\omega}$ remains the same after the change in $\underline{R}_D$.

D-ZLB) does not impact the first term, which is the marginal profit of a bank without limited liability. But through its effect on the cutoff, the D-ZLB does impact the limited liability subsidy. It is useful to look at the change in the marginal subsidy (for a given loan rate) when the bank-failure probability decreases:

$$\left.\frac{\partial MS}{\partial \bar{\omega}}\right|_{\hat{R}_L} = [(\hat{R}_L(1-\bar{\omega}) - R)L'(\hat{R}_L) + (1-\bar{\omega})L(\hat{R}_L)]f(\bar{\omega}) > 0. \tag{20}$$

Thus, a higher failure cutoff ceteris paribus increases the marginal subsidy. As (20) shows, this is because the marginal profit of raising the loan rate is positive at the failure cutoff.[20] Intuitively, as in Bahaj and Malherbe (2020), a higher failure cutoff $\bar{\omega}$ – i.e. a lower bank failure probability – shifts this positive marginal profit from the deposit insurance agency to the bank's shareholders, thereby increasing the total marginal profit of raising the loan rate for the bank's shareholders. Recall that the D-ZLB increases the failure probability of risky banks when it is binding (Corollary 1). As just seen, this decreases the marginal limited-liability subsidy that the bank receives (as the D-ZLB shifts positive marginal profits at the failure cutoff from the bank to the deposit insurance agency) and thereby decreases the marginal profits of lending for the bank's shareholders. But a bank that has market power in lending will raise the loan rate as long as the marginal profits of doing so is positive. In consequence, as Figure 3 illustrates, the optimal loan rate is lower, since marginal profits from raising the loan rate are positive to the left of the optimal loan rate.

3 The Amplification Mechanism Vis-à-Vis The Equity Erosion Channel

This section presents a quantitative, dynamic model to assess the relative importance of the risk-appetite channel emphasized in the previous section vis-a-vis the equity-erosion channel of e.g. Abadi et al. (2023). In the static model of the previous section, the equity-erosion channel was mute, since banks were not subject to capital requirements. In contrast, capital requirements and equity dynamics are now introduced, allowing the

[20] The sign of (20) follows since, by Eq. (18), $\Pi'(\hat{R}_L) = 0$ implies $sign[\hat{R}_L L'(\hat{R}_L) + L(\hat{R}_L)] = sign[RL'(\hat{R}_L)]$. Since $L'(\cdot) < 0$ the integrand in (18) is increasing in ω. But then for the integral in (18) to evaluate to zero, the integrand must be positive at the upper integration bound $\bar{\omega}$.

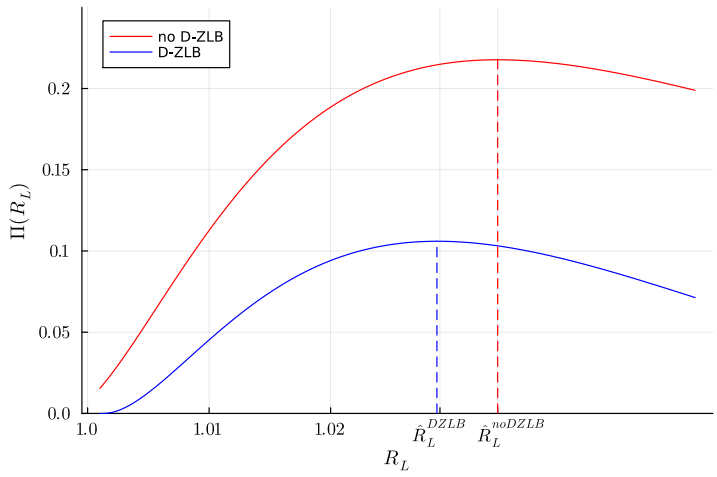


Figure 3: *Bank profits as a function of loan rates with and without a D-ZLB*

model to quantify the relative importance of the erosion of equity due to losses on deposits. The Online Appendix presents several variations and extensions of the model.

3.1 Quantitative Model

Set-Up. In the spirit of Mendicino et al. (2026), I assume that the economy consists of a unit continuum of islands k. On each island k there is an incumbent bank and a unit continuum of one-period firms i. The bank on island k can only lend to firms on its island and therefore has a local monopoly, but competes with all other banks in a monopolistic fashion in deposit markets. Banks also have access to a safe asset S remunerated at the policy rate R_t, which is common to all islands. Whenever the incumbent bank on an island fails it is immediately replaced with a new bank, which chooses its equity level endogenously, as in Corbae and D'Erasmo (2021).[21] In continuation, all model ingredients will be described in detail.

Monetary Policy. Monetary policy is taken as exogenous, and follows a two-state Markov chain, with states denoted by $s_t \in \{P, N\}$. The transition probability from state i to state j is given by q_{ij}. In state P, the central bank sets positive monetary policy rates $R(P) > 1$, while in state N the central bank lowers the monetary policy rate into negative territory $R(N) < 1$. State N will be associated with a recession caused by low firm productivity, as will be described below.

[21] As Corbae and D'Erasmo (2021) assume for "big banks", it is assumed that there is a cost to entering island k that is sufficiently high that no new banks enter as long as there is an incumbent bank.

Deposit Demand. Deposit markets are identical to the stylized model of Section 2. Deposits are perfectly insured by the government, and banks compete in a monopolistic fashion for deposits, with aggregate deposit rate R_{Dt}:

$$D(R_{Dkt}, R_{Dt}) = \left(\frac{R_{Dkt}}{R_{Dt}}\right)^{-\epsilon_D} \bar{D}, \ \epsilon_D < -1.$$

Monopolistic competition in deposit markets is commonly assumed in quantitative models studying the effects of the deposit zero lower bound, e.g. Abadi et al. (2023), Eggertsson et al. (2024) and Ulate (2021). By assuming a flat deposit demand, I do not take a stance on the debate about the deposit channel of monetary policy, that is whether bank deposits are decreasing in the policy rate or not (Drechsler et al., 2017; Begenau and Stafford, 2023; Repullo, 2020a).[22]

The model does not distinguish between deposit rates and account fees. Accordingly, the deposit rate can be interpreted as the net return on deposits inclusive of fees. The relevant lower bound is therefore not a lower bound on posted deposit rates per se, but on the overall return that households receive on deposits. Such a lower bound can be microfounded by assuming that households can substitute into cash whenever the net return on deposits falls sufficiently below the return on cash, which may itself be slightly negative because of storage and handling costs. Empirically, this interpretation appears consistent with evidence that banks' ability to offset the compression of deposit spreads through higher fees has been limited (Heider et al., 2019; Michaelis, 2022; Balloch and Koby, 2023).[23]

The Online Appendix extends the model to allow for imperfectly insured deposits and evaluates the quantitative importance of the perfect-insurance assumption. The results suggest that this assumption has only a limited effect on the model's main quantitative implications.

[22] While Drechsler et al. (2017) present evidence for the US that bank deposits are decreasing in the Fed funds rate, Begenau and Stafford (2023) challenge their identification strategy and find a statistically weak relationship between aggregate deposits and the Fed funds rate, and an increasing relationship between deposits and the Fed funds rate for the largest 10% of banks.

[23] For Germany, Michaelis (2022) finds an increase in fee income of 0.08 cents per euro of deposits during the negative interest rate period. Considering that the trough during this period was -50bp, banks were still incurring large losses in deposit taking. Heider et al. (2019) conclude that European banks highly exposed to negative interest rates did not raise fees vis-á-vis their peers. Balloch and Koby (2023) come to the same conclusion for Japanese banks for the prolonged period of low interest rates starting in 2000.

Firms: Production & Loan Demand. Loan demand and the loan default distribution are microfounded from the problem of firms that require bank loans for production.

At time t, firms learn about their state-dependent productivity $A(s_t)$, common to all firms on all islands. The representative firm i on island k requires bank loans to acquire production factors K_{ikt}, which depreciate at rate δ. The firm is subject to a binary failure shock $d_{ikt} \in \{0, 1\}$. When it is unsuccessful, the firm loses a share $\lambda > \delta$ of its production factors and defaults on its loan. The representative firm's revenue at time $t+1$ is then:

$$Y_{ikt+1} = (1 - d_{ikt+1})[(1 - \delta)K_{ikt} + A(s_t)K_{ikt}^{\alpha}] + d_{ikt+1}(1 - \lambda)K_{ikt} \quad \lambda, \alpha \in [0, 1]. \tag{21}$$

The bank on island k charges a gross interest rate $R_{Lkt} = 1 + r_{Lkt}$ for loans, and firms enjoy limited liability. Hence, the representative firm maximizes:

$$\max_{K_{ikt}} Pr(d_{ikt+1} = 0)[(1 - \delta)K_{ikt} + A(s_t)K_{ikt}^{\alpha} - R_{Lkt}K_{ikt}], \tag{22}$$

and the FOC is

$$(1 - \delta) + \alpha A(s_t)K_{ikt}^{\alpha-1} \stackrel{!}{=} R_{Lkt}. \tag{23}$$

implying a state-dependent loan demand of firm i on island k:

$$L_{ikt} = K_{ikt} = \left(\frac{R_{Lkt} - (1 - \delta)}{\alpha A(s_t)}\right)^{\frac{1}{\alpha-1}}, \tag{24}$$

in turn implying that total loan demand on island k is $L(R_{Lkt}, s_t) = \int_0^1 K_{ikt}dk = \left(\frac{R_{Lkt}-(1-\delta)}{\alpha A(s_t)}\right)^{\frac{1}{\alpha-1}}$, by symmetry. So, state-dependent loan-demand is a quantitatively relevant feature that allows the model to produce realistic loan-to-deposit ratios.

I turn, loan defaults follow the Vasicek (2002) single risk factor model, in which there is a common shock $z_{kt} \sim N(0, 1)$ to all firms on island k and an idiosyncratic shock $\epsilon_{ik} \sim N(0, 1)$ that realize at t, so:

$$d_{ikt} = \{\varsigma + \sqrt{\rho}z_{kt} + \sqrt{1 - \rho}\epsilon_{ikt} > 0\}, \tag{25}$$

where ς is a financial vulnerability parameter pinning down the unconditional default probability of all firms. As discussed in Gordy (2003), this setup is the model underlying

the Basel capital requirements, which banks in the model are subject to. As shown in Vasicek (2002), the cdf of the default rate ω is then given by (14).

The Online Appendix explores extensions of the model with (i) exogenously time-varying financial vulnerability, interpreted as reflecting financial crises and tranquil times, as well as (ii) firm default probabilities that increase with the loan rate.

Banks. As before, banks raise insured deposits D, invest in risky loans L and a safe asset S that is remunerated at the policy rate.[24] Different to the static illustrating model in Section 2, banks are now subject to a capital requirement $\gamma L_{kt} \leq E_{kt}$, where E_{kt} denotes the equity of the bank operating on island k after paying out dividends and any equity injections. This is the main difference between the quantitative and the illustrating model, giving rise to an equity-erosion channel as in Abadi et al. (2023) and Ulate (2021), through which banks are forced to reduce lending when losses on deposits are sufficiently large. Thus, the representative bank's balance sheet is now:

$$E_{kt} + D_{kt} = L_{kt} + S_{kt}. \tag{26}$$

Every period, banks decide the amount $\nu_{kt} > 0$ to be paid out as dividends, and how much new equity I_{kt} to raise. The representative bank can raise equity at a quadratic cost χI_{kt}^2, similar to Corbae and D'Erasmo (2021), while dividend payouts are frictionless. Hence, equity dynamics are completely endogenous here – different to both Abadi et al. (2023) and Ulate (2021), where the dividend payout ratio is fixed and exogenous.

As just discussed, an island-specific fraction ω_{kt} of any bank's loan portfolio defaults, with a loss given default of λ, so the ex-post realized return on the loan portfolio of the bank operating on island k is:

$$\tilde{R}_{Lkt}(\omega_{kt}) = (1 - \omega_{kt})R_{Lkt-1} + \omega_{kt}(1 - \lambda) \tag{27}$$

The bank operating on island k fails if its loan repayments are insufficient to repay its deposit obligations in full, i.e. if the loan default rate ω_k exceeds the threshold $\tilde{\omega}_{kt}$

[24] An extension of the model that allows for additional non-deposit funding will be considered in Section A.6.

implicitly defined by:

$$\tilde{R}_{Lkt}(\tilde{\omega}_{kt})L(R_{Lkt-1}, s_{t-1}) - R_{Dkt-1}\bar{D} + R(s_{t-1})S_{kt-1} = 0, \tag{28}$$

in which case, the bank is closed down forever and the bank's assets are repossessed by a deposit insurance agency, discussed further below. As mentioned above, it is assumed that upon the failure of the incumbent bank on island k, a new bank enters on island k immediately.

Next, I assume that banks cannot charge gross deposit rates below $\underline{R}_D$. In the baseline model, I set $\underline{R}_D = 1$ in line with the ample empirical evidence for a zero lower bound on deposit rates (Eggertsson et al., 2024; Basten and Mariathasan, 2023; Hong and Kandrac, 2021). Since the only difference between conventional monetary policy rates and negative interest rate policy is that in the latter case the D-ZLB constraint is binding, the quantitative analysis will focus on the comparison of the baseline scenario vis-a-vis an alternative scenario in which banks are not subject to a D-ZLB. The difference between the scenarios is informative about the effects of negative interest rate policy vis-a-vis conventional monetary policy, as without the D-ZLB, a policy rate cut from positive to negative rates is like any other cut.

Banks are managed in the interest of their shareholders, who enjoy limited liability and discount the future at a rate $\beta(s_t)$, a function of the risk free rate $R(s_t)$. For quantitative purposes, similar to Repullo and Suarez (2012), I assume that bank shareholders discount the future at a rate higher than $R(s_t)$: $\beta(s_t) = \frac{1}{R(s_t)+\Delta_E}$, which reflects a constant, exogenous excess cost of equity.[25]

The bank's value function can be written as $V(s, \tilde{E}_k)$, where time subscripts are omitted and $\tilde{E}_k$ denotes the equity of the bank currently operating on island k before dividend payments and equity injections. The individual bank takes the aggregate deposit rate as given. Since, as shown below, the equilibrium deposit rate is a function only of the exogenous aggregate state s, I write it as $R_D(s)$.

By limited liability of shareholders, if the incumbent bank fails, its shareholders receive

[25] Δ_E reflects a differential cost of equity as observed in reality. As discussed in Repullo and Suarez (2012), such a cost might reflect costs of monitoring managers incurred by shareholders, a discount for lack of liquidity of equity stakes and a risk-related component. I follow Repullo and Suarez (2012) in abstracting from changes in the risk-premium component of the cost of equity due to changes in bank's leverage (Admati et al., 2010).

no continuation value. This is indicated by the binary variable ($\mathcal{D}'_k$), which equals one if the incumbent bank fails and zero otherwise.

Thus, the representative bank's problem is:

$$V(s,\tilde{E}_k) = \max_{\nu_k\in[0,\tilde{E}_k],I_k\geq 0,R_{Lk},R_{Dk},S_k\geq 0} \nu_k - I_k - \chi I_k^2 + \beta(s)\mathbb{E}(1-\mathcal{D}'_k)V(s',\tilde{E}'_k)$$

$$\begin{aligned}
\text{subject to } & \gamma L(R_{Lk},s) \leq \tilde{E}_k - \nu_k + I_k, \\
& L(R_{Lk},s) + S_k = D(R_{Dk},R_D(s)) + \tilde{E}_k - \nu_k + I_k, \\
& R_{Dk} \geq \underline{R}_D, \\
& \tilde{E}'_k = \tilde{R}'_{Lk}(\omega'_k)L(R_{Lk},s) - R_{Dk}D(R_{Dk},R_D(s)) + R(s)S_k, \\
& \mathcal{D}'_k = \begin{cases} 1 \text{ if } \tilde{E}'_k < 0, \\ 0 \text{ else.} \end{cases}
\end{aligned} \tag{29}$$

Thus, while the bank that enters island k upon failure of the incumbent bank enters with $\tilde{E}_k = 0$, its initial equity injection I_k is determined endogenously and depends on the cost parameter χ.

As in the static model of Section 2 the representative bank sets the same deposit rate independently of the loan rate. Specifically, the deposit rate FOC is:

$$D(R_{Dk},R_D(s)) + (R_{Dk} - R(s))\frac{\partial D(R_{Dk},R_D(s))}{\partial R_{Dk}} = \lambda^{LB}, \tag{30}$$

where the multiplier on the deposit lower bound is denoted by λ^{LB}, which will be binding in state N since $R(N) < 1$ and non-binding in state P. Thus, all banks will offer the same deposit rate, such that

$$R_{Dkt} = R_{Dt} = \begin{cases} 1 \text{ if } s_t = N \\ \frac{\epsilon_D}{\epsilon_D - 1}R_t \text{ if } s_t = P, \end{cases}$$

and further $D(R_{Dkt},R_{Dt}) = \bar{D}$. In state N, banks therefore face losses on deposits, which makes them ceteris paribus (a) more likely to fail and (b) erodes their equity over time. Naturally, this impacts the likelihood that a bank will be constrained by the capital requirement, which is the channel emphasized by Abadi et al. (2023) in a model abstracting from any risk. The simple model in Section 2 predicts that when banks are

risky the D-ZLB impacts the loan supply decision even for unconstrained banks. The following sections will confirm that this conclusion still holds here.

Equilibrium. The aggregate state of the economy is given by the monetary-policy state s_t and the cross-sectional distribution of bank equity, denoted by Γ_t. The distribution Γ_t is an endogenous aggregate state variable because individual banks' equity evolves endogenously and hence determines the cross-sectional distribution of equity across islands. Although Γ_t is part of the aggregate state, it does not enter the individual bank's optimization problem, either directly or through the equilibrium deposit rate. As shown above, the latter is determined solely by s_t. An equilibrium consists of individual bank policy functions and a law of motion for the distribution of bank equity such that the policy functions solve the bank's problem given s_t and the equilibrium deposit rate $R_D(s_t)$.[26] Aggregate quantities are then obtained by integrating the corresponding bank-level variables over the distribution Γ_t, which is equivalent to integrating over the set of islands k, given that there is a single bank per island. Thus, aggregate lending is:

$$L_t = \int_0^1 L_{kt} dk, \tag{31}$$

and the aggregate loan rate is given by:

$$R_{Lt} = \int_0^1 R_{Lkt} dk. \tag{32}$$

Deposit Insurance Agency. To allow an assessment of the cost of deposit insurance due to the D-ZLB, I explicitly model a deposit insurance agency. This agency repossesses the assets of a bankrupt bank and uses the proceeds as well as government funds T_t to repay depositors. I assume that repossession of the loan portfolio is costly, such that the DIA only receives $(1-\mu)$ of the loan repayments of failing banks. The amount of government funds the DIA needs to repay depositors is then:

$$\begin{aligned} T_t = \int_0^1 \mathbf{1}(\omega_{kt} > \tilde{\omega}_{kt}) \big\{ & R_{Dt-1}\bar{D} - R_{t-1}S_{kt-1} \\ & -(1-\mu)\left[(1-\omega_{kt})R_{Lkt-1} + \omega_{kt}(1-\lambda)\right] L(R_{Lkt-1}, s_{t-1}) \big\}\, dk. \end{aligned} \tag{33}$$

[26] Given the individual policy functions, the law of motion of the distribution of equity Γ_t evolves according to the law of motion of individual incumbent banks' equity and their failure.

3.2 Calibration

The model is calibrated at yearly frequency for Germany, which has a banking sector with a particularly high deposit-to-loan ratio.[27]

To calibrate the states of the economy and the transition probabilities, I use data on the deposit facility rate and the main refinancing operations (MRO) rate set by the ECB between Q1 1999 and Q3 2023. Before the financial crisis, the main policy rate steering money markets in the Euro Area was the minimum bid rate of the main refinancing operations (see Hartmann and Smets, 2018). From October 2008 onwards, this role was played by the deposit facility rate (Hartmann and Smets, 2018).

I then split the time series of the relevant gross policy rate in two regimes: a negative monetary policy rate regime (state N), with the relevant rate below 0% per annum, and a positive monetary policy rate regime (state P) with the relevant rate above 0% per annum. Let the resulting time-series be denoted R_t^{obs}. The average policy rates in the two regimes are $R(P) = 1.0237$ (that is, 2.37% per annum) and $R(N) = 0.9969$ (that is, -0.31% per annum). I estimate the transition probabilities of the Markov chain:

$$Pr(R_t = R(j)|R_{t-1} = R(i)) = q_{ij} \ , i, j \in \{N, P\}, \tag{34}$$

using the Maximum Likelihood estimator, given by

$$q_{ij} = \frac{n_{ij}}{n_{iP} + n_{iN}},$$

where $n_{ij} = \sum_t \mathbf{1}\{R_t^{obs} = R(j)\}\mathbf{1}\{R_{t-1}^{obs} = R(i)\}$ (see e.g. Norris, 1998). The results are: $q_{PP} = 0.933$, $q_{PN} = 0.067$, $q_{NP} = 0.09$, $q_{NN} = 0.91$.

To calibrate the discount factors of banks, I rely on estimates of the cost of bank equity from European Central Bank (2015) and Fernández and Mencía (2020). Given this evidence, a value of 5% per annum appears a reasonable choice.[28] I therefore set the

[27] An alternative calibration for the euro area, where the average bank had a deposit-to-loan ratio well below one during the NIRP period, is presented in the Online Appendix in a robustness exercise in a slightly extended model that allows for the non-deposit debt funding necessary to match such low deposit-to-loan ratios (Section A.6).

[28] European Central Bank (2015) reports a decomposition of the cost of bank equity into the real risk free interest rate, inflation expectations and the equity risk premium based on estimates of a dividend discount model and the CAPM for 33 European banks listed in the EURO STOXX index. Prior to the financial crisis, the excess cost of bank equity (relative to the risk free real interest rate) was relatively stable between 3% and 7%, with little differences in the total cost of equity between the four largest economies (Germany, France, Italy, Spain) for which country-specific estimates are reported. Using a

excess cost of bank capital to $\Delta_E = 0.05$. Further, I set the depreciation rate to $\delta = 0.1$ and the capital requirement to the Basel II capital charge $\gamma = 0.08$ as is standard. I follow Repullo and Suarez (2012) in setting the loss-given-default parameter $\lambda = 0.45$. As discussed therein, this was the value calibrated for uncollateralized corporate loans in the Internal Rating Based Approach introduced in the Basel II framework. Next, I set $\epsilon_D = -100$ to imply a deposit spread of 1% per annum for banks in the positive monetary policy state, similar to the calibration of Abadi et al. (2023). The flat aggregate deposit supply is normalized to $\bar{D} = 50$, which is numerically advantageous. Next, I set the firm's unconditional default probability (given by $\Phi(\varsigma)$, where $\Phi(\cdot)$ is the cdf of the standard normal distribution) to 2.7%, based on euro-area data reported in Mendicino et al. (2026). Lastly I follow Mendicino et al. (2026) in setting the repossession cost of assets of a bankrupt bank to 30%, i.e. $\mu = 0.3$.

The remaining parameters $\alpha, \rho, A(N), A(P)$ and χ are estimated using the Simulated Method of Moments. To pin down α, I target a loan rate spread of 2.56% (calculated from average loan rates in Germany to non-financial corporations published by the Bundesbank). For the correlation between firm defaults ρ, I target the mean bank-failure probability of the euro area as a whole (taken from Mendicino et al. 2026). The state-dependent productivity of firms $A(N), A(P)$ is pinned down by the mean deposit-to-loan ratio in both states (defined as above based on the policy rate), calculated from data on German banks' balance sheets available from the ECB.[29] Lastly, for the cost of equity injections χ I target the ratio of equity injections to total equity calculated from Eurostat Sectoral National Accounts data.[30]

The estimated values are $A(P) = 0.197$, $A(N) = 0.157$, $\alpha = 0.95$, $\rho = 0.263$ and

similar methodology, Fernández and Mencía (2020) report similar levels of the cost of bank equity (in excess of the risk free rate) for October 2020 (after a covid-related spike) as in early 2007. Abstracting from market turmoils due to the financial crisis, sovereign debt crisis, Brexit and covid between 2008 and 2020, a 5% excess cost of bank equity therefore seems reasonable in the context of this model.

[29]Deposits: sum of Deposit liabilities vis-a-vis euro area households, non-financial corporations (NFCs) and Insurance and Pension Fund (ICPFs) reported by MFIs excl. ESCB in Germany (BSI.M.DE.N.A.L20.A.1.U2.2250.Z01.E + BSI.M.DE.N.A.L20.A.1.U2.2220.Z01.E + BSI.M.DE.N.A.L20.A.1.U2.2240.Z01.E); Loans: Loans vis-a-vis euro area Non-MFIs excluding general government reported by MFIs excl. ESCB (BSI.M.DE.N.A.A20.A.1.U2.2200.Z01.E).

[30]The aggregate equity injections are approximated by the financial transactions account for equity of financial corporations in Germany. Since this includes stock repurchases, the resulting number is a lower bound for equity injections. The data for aggregate equity is: Eurostat dataset NASQ_10_F_BS (quarterly financial balance sheets), sector S12 (financial corporations), Liabilities, item F51 (equity). The data for injections is: Eurostat dataset NASQ_10_NF_TR (quarterly financial transactions), sector S12 (financial corporations), Liabilities, item F51 (equity).

Table 1: *Model Fit*

Moment	Data	Model
Mean Loan Spread	2.56	2.56
Mean Bank Default	0.66	0.61
L/D (State P)	1.06	1.05
L/D (State N)	0.92	0.93
Equity Injections to Total Equity	> 0.3	0.34

Note: The loan spread is calculated as the difference of corporate lending rates as reported by the Bundesbank (2003-2023, excluding 2008/2009) and the relevant policy rate; bank-failure probability from Mendicino et al. (2026) for the euro area; Deposit-to-Loan Ratio for Germany from ECB BIS (1999Q1-2023Q3), see Footnote 29; Equity injections are calculated from National Accounts data from Eurostat, see Footnote 30.

$\chi = 0.7$. The data moments and model moments are summarized in Table 1. The Online Appendix conducts robustness checks with alternative parameters.

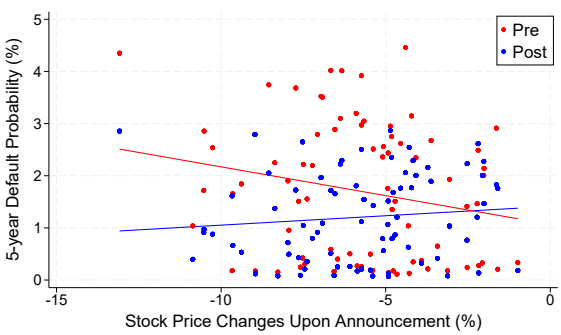
3.3 Testable Implication: Equity and Riskiness.

The model implies an endogenous, state-dependent relationship between bank equity and default risk. With a fixed deposit base and capital requirements, equity determines lending capacity and hence exposure to credit risk relative to the deterministic payoff from deposits when rates are positive. Meanwhile, under negative rates, losses on deposits are larger relative to equity for less-capitalized banks. Thus, bank riskiness tends to increase in equity in state P, and decrease in equity in state N. To assess the plausibility of this, I compare the model's predictions with Japanese bank-level data from the publicly available replication package of Hong and Kandrac (2021).[31] In the model, the percentage change in bank value when moving from positive to negative rates varies monotonically with pre-dividend equity $\tilde{E}_k$ (Appendix Figure 8). Since a bank's stock price equals its value divided by the number of shares outstanding, this percentage change corresponds to the stock-price response, holding the number of shares fixed. I therefore compare the relationship between the model-implied change in bank value and one-year failure probabilities with that between stock-price changes in the 40-minute window around the announcement of NIRP and five-year default probabilities in the Japanese data.[32] Despite

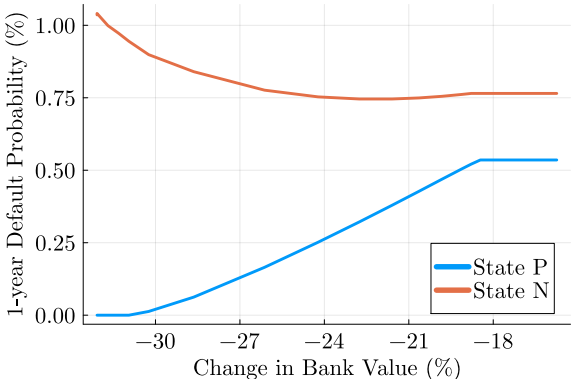
[31] I do not have access to comparable data on German banks' default probabilities.

[32] Hong and Kandrac (2021) use stock-price changes in a 40-min window around the announcement of negative interest rates as a measure of exposure to negative interest rates, and study the impact on 5-year default probabilities from the Kamakura corporation.

differences in country and default-probability horizon, the qualitative consistency of the relationships supports the plausibility of the model's implications for the joint distribution of bank safety and equity.



(a) *Data from Hong and Kandrac (2021) and own calculations*



(b) *Model*

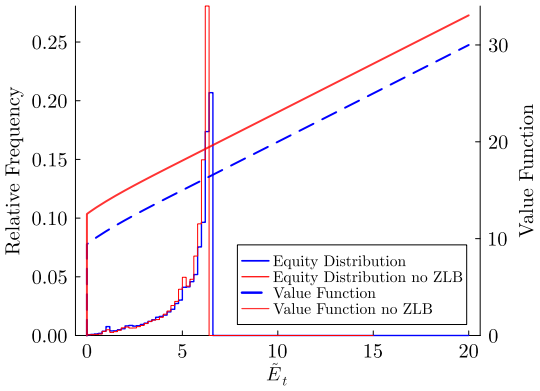
Figure 4: *Default Probability and Price Impact: Comparison Model and Data*

In the data from Hong and Kandrac (2021), the stock price change is calculated in a 40min window around the announcement of a monetary policy cut to -10bps. In the model, the change in bank value is the change in the value function $V(s, \tilde{E})$ when the exogenous state of the economy s changes from P to N: $100 \times \left[V(N, \tilde{E}) - V(P, \tilde{E})\right] / V(P, \tilde{E})$.

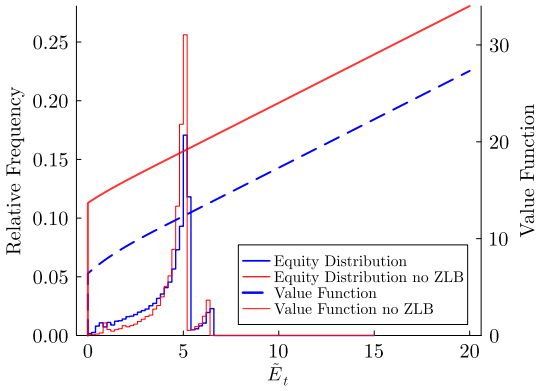
3.4 Quantitative Results

To quantify how negative interest rate policy differs from conventional monetary policy, I compare the baseline economy, in which banks are subject to a deposit zero lower bound (D-ZLB) ($\underline{R}_D = 1$), with a counterfactual economy in which deposit rates can adjust freely ($\underline{R}_D \to -\infty$). Through the lens of the model, the D-ZLB is the only feature that distinguishes negative interest rate policy from conventional monetary policy. Differences between the two economies therefore isolate the additional effects of negative interest rate policy arising from the D-ZLB beyond those expected from conventional monetary policy.

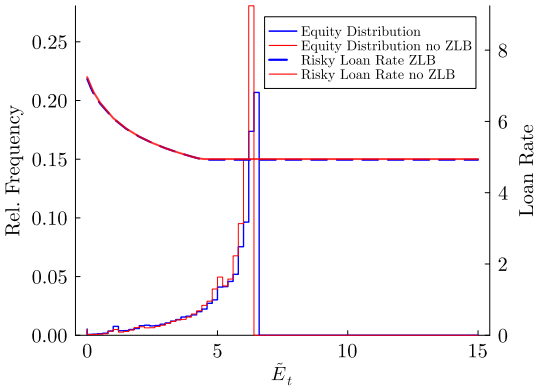
Figure 5 depicts the value functions and policy functions of the representative bank (that is, loan rates, dividends and equity injections as a function of pre-dividend equity $\tilde{E}_{kt}$), alongside the distribution of $\tilde{E}$, obtained from simulating 10,000 independent islands over 1,000 periods. Three observations stand out. First, unconstrained banks charge lower loan rates when the D-ZLB is present, reflecting the risk-appetite channel identified in the illustrating model. Second, banks respond to the losses in deposit-taking generated by



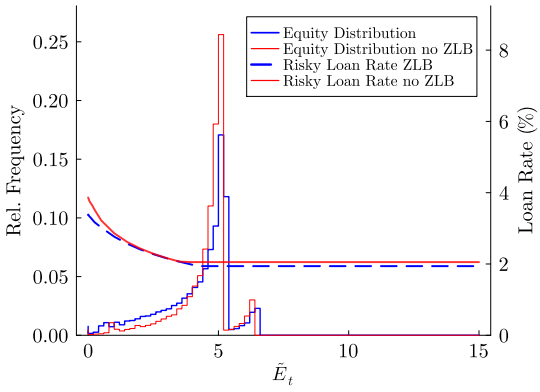
(a) *Value Function* $s_t = P$



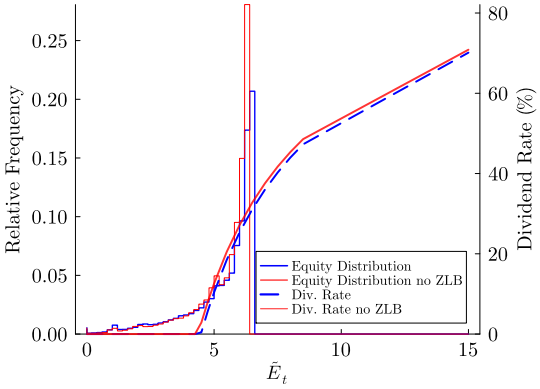
(b) *Value Function* $s_t = N$



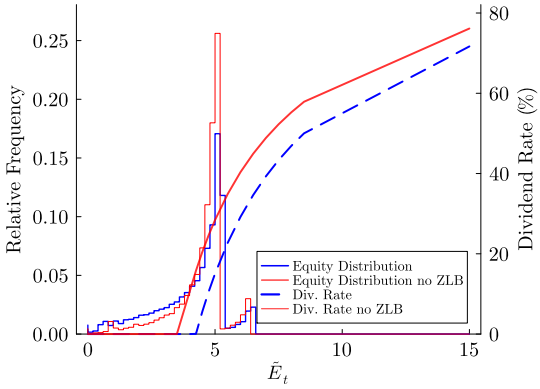
(c) *Loan Rate* $s_t = P$



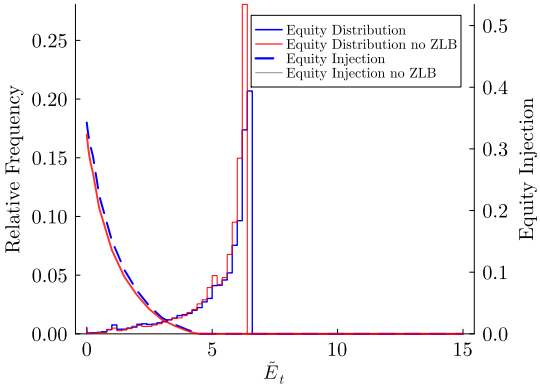
(d) *Loan Rate* $s_t = N$



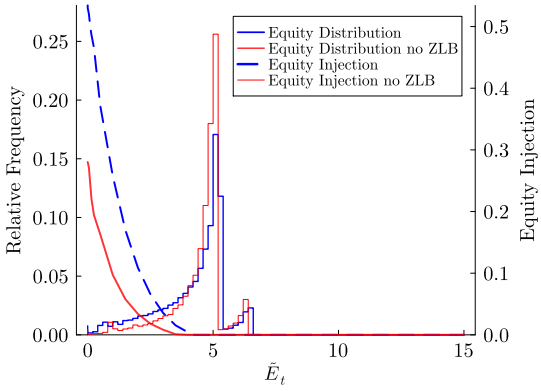
(e) *Dividend Ratio* $s_t = P$



(f) *Dividend Ratio* $s_t = N$



(g) *Equity Injection* $s_t = P$



(h) *Equity Injection* $s_t = N$

Figure 5: *Policy Functions and Equity Distribution*

the D-ZLB by retaining more earnings and raising equity more aggressively. Relative to the no-D-ZLB economy, equity injections occur at higher levels of $\tilde{E}$. In turn, for equity-levels at which the bank stops raising equity and instead pays dividends, the dividend payouts are lower relative to the no-D-ZLB economy. Third, These adjustments partially, but not fully, mitigate the erosion of bank capital and allow banks to sustain larger loan portfolios. As a result, the probability of banks having low equity levels, which force them to restrict lending due to a binding capital requirement, increases relative to the no-D-ZLB economy – in other words, the left tail of the equity distribution is notably thicker when banks are subject to a D-ZLB and policy rates are negative. Fourth, the D-ZLB affects bank decisions even when policy rates are positive. The possibility of future negative rates lowers franchise values and increases risk-taking incentives, generating a charter-value effect akin to that in Repullo (2004). Unsurprisingly, these effects are substantially stronger when policy rates are actually negative.

The results of simulating the model are summarized in Table 2. The first two columns present the results of the baseline model in which banks are subject to a binding D-ZLB in state N (i.e. $\underline{R}_D = 1$) by state, while columns 3 and 4 present the results of the alternative scenario in which banks are not subject to a D-ZLB. The table shows that the quantitative effects of the D-ZLB are substantial. When the policy rate is -0.31%, unconstrained banks charge loan rates that are 11 basis points lower than in the counterfactual economy without a D-ZLB. At the same time, the share of capital-constrained banks rises from 15% to 38%, reflecting the equity-erosion channel. Despite this deterioration in bank capitalization, the risk-appetite channel dominates quantitatively: aggregate loan supply is 9% higher under the D-ZLB.

The increase in lending is accompanied by a meaningful deterioration in bank solvency. Under negative policy rates, the D-ZLB raises the bank-failure probability by 30 basis points, from 0.47% to 0.77%. Thus, the binding D-ZLB increases bank failure risk by roughly 64% relative to the counterfactual economy. Nevertheless, the additional deposit insurance cost due to the D-ZLB is relatively small.

For the reasons discussed above, the impact of the D-ZLB continues to affect the economy in the positive monetary policy state P. However, the impact is quantitatively much smaller. Hence, relative to the counterfactual economy without D-ZLB, the transmission of the policy rate cut from positive to negative rates is stronger: a change from

Table 2: *Average Results By State*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	52.48	46.46	51.77	42.52
Loan Rate (unconst., %)	4.92	1.94	4.94	2.05
Avg. Loan Rate (const. %)	5.34	2.27	5.36	2.39
Avg. Loan Rate (%)	5.01	2.07	5.01	2.11
Share Constrained (%)	19.52	37.98	16.92	15.19
Bank Failure Prob. (%)	0.54	0.77	0.52	0.47
Deposit Insurance Costs	0.08	0.11	0.08	0.06
Deposit Insurance Costs (Unconditional)		0.092		0.076
Loan Volume (Unconditional)		50.657		48.966
Loan Volume Difference (State N - P)		-6.02		-9.25
D-ZLB Bank Risk Effect	0.02	0.3		

Note: This table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

state P to state N – which in the model implies a drop in firm productivity and hence a downward shift in loan demand – is associated with a significantly lower decrease in loan volumes due to the D-ZLB.

To shed more light on the heterogeneity of the D-ZLB effect on the cross-section of loan supply, Figure 6 illustrates the cross-sectional heterogeneity of the D-ZLB effect. The figure plots the distribution of differences in loan supply between the baseline and counterfactual economies across islands. In both states, the distribution exhibits a mode to the right of zero, reflecting the risk-appetite channel, and a left tail generated by banks whose lending is constrained by low equity. The left tail is considerably thicker in the negative-rate state N, where equity erosion is strongest. In state P, loan supply is lower under the D-ZLB in only 6% of islands and higher in 94%. In state N, the corresponding shares are 25% and 75%, respectively. The next subsection examines how this cross-sectional pattern evolves over the duration of a negative-rate episode.

3.5 Dynamics: The D-ZLB Effect Over Time

In the model, periods of negative interest rates coincide with periods of low productivity, $A(s)$. The results presented so far average over time at the baseline policy rate of

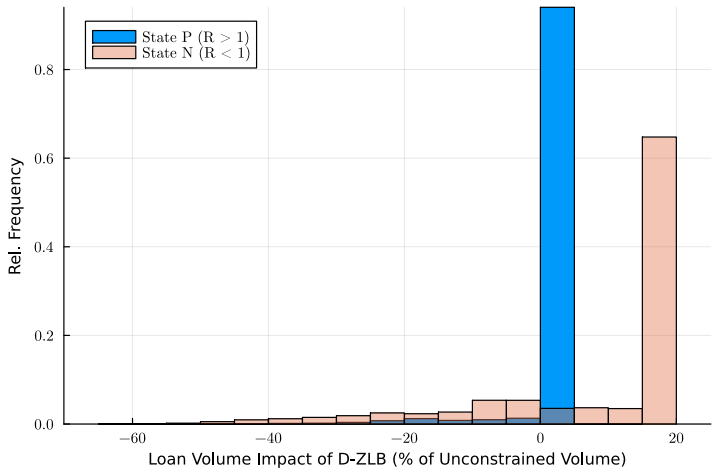


Figure 6: *Histogram Loan Supply Differences*

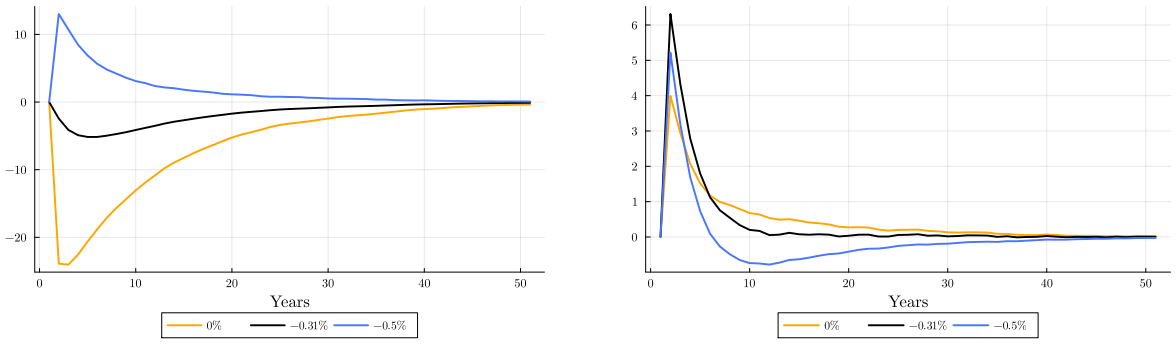
Note: This figure depicts the distribution of differences $L_{kt}^{DZLB} - L_{kt}^{noDZLB}$ in loan supply in the scenario in which banks are subject to a D-ZLB and the scenario in which they are not subject to a D-ZLB, by aggregate state. The sequence of loan default rates ω_{kt} on a given island k is kept constant across simulations.

-0.31% in the low-productivity state N. I now examine how the effects of the D-ZLB evolve over time following a productivity decline and how they depend on the depth of the monetary policy easing. Specifically, I consider policy rates of 0%, -0.31% (the baseline), and -0.5% in state N.

Figure 7 plots impulse responses following a transition from the high-productivity state P to the low-productivity state N, under each of the three policy-rate scenarios. The productivity shock occurs in period $t = 1$, after which the economy follows the Markov-chain dynamics: in each subsequent period, there is approximately a 10% probability of returning to state P.

Because banks differ in their initial equity levels and subsequent loan default realizations, their individual responses to the shock are heterogeneous. The impulse responses therefore report averages across banks, which coincide with aggregate loan supply.[33] Panel (a) shows the impulse responses of aggregate loan supply. While the productivity shock shifts down loan demand, monetary policy easing is expansionary. Loan supply still falls following the productivity decline under the 0% and -0.3% policy-rate scenarios, but the decline is smaller at -0.3%, reflecting the stronger expansionary effect of monetary easing. At -0.5%, the impulse response is positive: the stronger monetary stimulus more than offsets the contractionary effect of lower productivity, leading to an increase in aggregate

[33] Aggregate loan supply is given by $L_t = \int_0^1 L_{kt}\, dk$, and hence coincides with average loan supply.



(a) *Mean Loan IRF* **(b)** *Difference Loans IRFs (D-ZLB - no D-ZLB)*

Figure 7: *Evolution After Temporary Productivity Change From $A(P)$ To $A(N)$*

Note: This figure depicts the average evolution across islands k of selected variables after a temporary change from state P to state N for varying policy rates in state N. The policy rate in state P, as well as the sequence of loan default rates ω_{kt} is kept constant across simulations.

lending.

Panel (b) plots the difference between aggregate loan supply in the economy with a D-ZLB and in the counterfactual economy without one. The difference is initially positive for all three policy-rate scenarios, indicating that monetary easing stimulates lending more strongly when the D-ZLB binds. This amplification is strongest immediately after the policy-rate cut and subsequently diminishes as the equity-erosion channel increasingly offsets the risk-appetite channel.

For policy rates of 0% and -0.3%, the D-ZLB continues to raise aggregate lending relative to the counterfactual as long as the economy remains in the negative-rate state. The results differ for the deeper cut to -0.5%: after approximately six years, the equity-erosion channel dominates, and lending becomes lower under the D-ZLB than in the counterfactual without it. These results are confirmed when considering a permanent transition to state N, instead of a temporary transition (see Figure 9 in the Appendix).

The average results by aggregate state for the alternative policy-rate scenarios are reported in Tables 8 and 9 in the Online Appendix. In all three cases, average aggregate loan supply is higher under the D-ZLB when policy rates are negative. Taken together, these results emphasize that even though the D-ZLB amplifies the expansionary effect of monetary easing on average, this amplification weakens over time and can eventually become less expansionary following a sufficiently deep cut.

3.5.1 Cuts Within Negative Territory

The analysis so far has focused on policy-rate cuts from a level at which the D-ZLB does not bind to one at which it does. Previous theoretical contributions, however, have also examined the effects of *marginal* policy-rate cuts within negative territory (Ulate, 2021; Abadi et al., 2023). Corollary 3 establishes that, under restrictive conditions, a marginal cut is more expansionary when the D-ZLB binds in the illustrating model, where banks face no capital constraints. I now investigate whether this result carries over to the quantitative model, in which equity erosion can constrain banks' lending.

I simulate additional, unexpected policy-rate cuts of 10 basis points within negative territory, similar to those implemented by the ECB between 2014 and 2022. The deepest policy rate considered is thus -0.5%. In each experiment, the economy first transitions to the low-productivity state N at $t = 1$, triggering an initial monetary easing. A further, unexpected 10-basis-point cut then occurs after one, five, or ten years at t_2, i.e. $t_2 \in \{2, 6, 11\}$. While banks assume that the probability of returning to state P remains unchanged at $q_{NP} = 0.09$, the simulated cut is permanent to emphasize its long-run effects.

Table 3 summarizes the findings. The signs indicate whether the additional policy-rate cut raises aggregate loan supply more (+) or less (−) under the D-ZLB than in the counterfactual without a D-ZLB; 0 indicates an approximately neutral difference. The table therefore captures the effect of the D-ZLB on the transmission of each cut, rather than the overall effect of the cut itself. Additionally, Figure 13 in the Appendix depicts the complete impulse responses and the D-ZLB effect over time – similar to the preceding analysis, all monetary policy cuts considered remain expansionary in the presence of the D-ZLB, as the figure shows.

Table 3 shows that the effect of the D-ZLB on the transmission of marginal cuts depends on both the level of the policy rate and the time elapsed since the initial easing. In the long run, the D-ZLB amplifies marginal cuts ending at up to -0.4%, at which point the effect is approximately neutral, and turns negative for the cut ending at -0.5%. In the short run, however, the results are more heterogeneous. Starting at -0.2%, the marginal cuts are less expansionary in the short run under the D-ZLB when the second cut occurs sufficiently long after the initial easing, while being more expansionary otherwise.

This short-run pattern reflects the interaction between deposit-taking losses due to

the D-ZLB and banks' ability to accumulate equity. Following the second monetary policy cut, banks need additional equity to support their increased desired loan supply (remember that productivity, and thus the loan demand curve, remains fixed at this second cut). This process is slower when banks incur losses in deposit-taking than when they continue to earn profits on deposits. Consequently, after a marginal monetary policy cut that is not too deep in negative territory, the D-ZLB can initially reduce lending vis-a-vis the counterfactual and yet increase it in the long run.

Overall, the marginal-cut experiments qualify the earlier results: within negative territory, even cuts that the D-ZLB amplifies in the long run may in the short run be less effective due to its equity-eroding effects, depending on the timing of the cuts.

Table 3: *Impact of D-ZLB on Loan Supply for Marginal MP Cuts*

Short or Long Run?	First cut to	Second cut to	at $t_2 = 2$	$t_2 = 6$	$t_2 = 11$
Short Run	-0.1%	-0.2%	+	+	+
Long Run	-0.1%	-0.2%	+	+	+
Short Run	-0.2%	-0.3%	+	+	-
Long Run	-0.2%	-0.3%	+	+	+
Short Run	-0.3%	-0.4%	+	-	-
Long Run	-0.3%	-0.4%	0	0	0
Short Run	-0.4%	-0.5%	+	-	-
Long Run	-0.4%	-0.5%	-	-	-

The signs indicate whether the additional policy-rate cut raises aggregate loan supply more (+) or less (−) under the D-ZLB than in the counterfactual without a D-ZLB; 0 *indicates an approximately neutral difference.*

3.6 Alternative Policy: Lower Capital Requirements Instead of Negative Interest Rates

As we have seen in the previous section, negative interest rates stimulate the economy at the expense of an increase in bank riskiness. A similar qualitative effect can be expected from a decrease in capital requirements in state N, i.e. in response to the negative productivity shock. In reality, such a decrease could for example be achieved by a release of countercyclical capital buffers.

Can such a decrease in capital requirements achieve a similar stimulation of the econ-

omy, at a lower increase in bank riskiness? To check this, I set the policy rate in state N to $R(N) = 1$ (i.e. a 0% per annum net policy rate) and decrease capital requirements in state N to (a) $\gamma(N) = 0.07$, (b) $\gamma(N) = 0.06$ and (c) $\gamma(N) = 0.05$. The results are presented in Table 4. While the decreases in the capital requirement lead to increasingly higher lending in state N, they fail to achieve a similar level of stimulation as the baseline negative policy rate of -0.31%. At the same time, the bank-failure probability increases beyond the baseline level when capital requirements are lowered to 5%, despite banks not facing losses in deposit taking due to the higher policy rate of 0%. Therefore, negative interest rate policy appears the better choice here overall.

Table 4: *Release of Capital Requirements*

Policy Rate (%)	-0.31	0.0	0.0	0.0
Capital Requirement (%, State P)	8.0	8.0	8.0	8.0
Capital Requirement (%, State N)	8.0	7.0	6.0	5.0
Loan Volume (P)	52.483	51.494	51.53	51.337
Loan Volume (N)	46.458	27.928	28.313	31.028
Bankruptcy Prob. (%, P)	0.539	0.518	0.52	0.536
Bankruptcy Prob. (%, N)	0.766	0.547	0.566	0.857
Deposit Insurance Costs (P)	0.085	0.082	0.083	0.082
Deposit Insurance Costs (N)	0.109	0.045	0.047	0.083

Note: This table reports the average aggregate loan volume, bank failure probability and deposit insurance cost by aggregate state $s_t \in \{P, N\}$ for different policy combinations in state N. The first column corresponds to the baseline.

3.7 Can Capital Requirements Complement Negative Interest Rates?

I next assess whether higher capital requirements (i) in state P can ameliorate the effects of the equity-erosion channel on lending in state N or (ii) in state N can ameliorate the increase in bank-failure probabilities due to the D-ZLB in state N. Table 5 presents the results.

A higher capital requirement in state P mimics a positive neutral countercyclical capital buffer (CCyB). As discussed in Behn et al. (2023), the rationale for such a buffer is to build up capital buffers in tranquil times – corresponding to state P in the model – in order to release them and thus increase the lending capacity of banks when an adverse

shock hits. To assess the role of such an increased buffer in state P, I increase the capital requirement in state P to (i) $\gamma(P) = 0.09$ and (ii) $\gamma(P) = 0.10$, i.e. a positive neutral CCyB of 1% or 2%, in line with those used in euro-area countries in reality (Behn et al., 2023). In the model, the positive neutral CCyB significantly decreases lending in state P, while leading to a small increase in lending and a small decrease in the bank-failure probability in state N compared to the baseline. The effects in state N are due to banks entering state N with a slightly higher level of equity, protecting them from losses in deposit taking and slightly decreasing the share of constrained banks in that state. However, the decrease in lending in state P implies that the higher capital requirements have a limited effect on the equity with which banks enter state N – thus, the increase in bank lending in state N when increasing capital requirements in state P is small. It is not clear that the quantitatively small improvements in state N justify the large decrease in bank lending in state P.

On the other hand, the higher capital requirements in state N show a clear trade-off between lower higher loan volumes and safer banks. For a capital requirement of 9.5%, the bank-failure probability is approximately as high as it would be in absence of the D-ZLB (compare Table 2), but the loan volume is lower. Hence, the central bank cannot mute the financial stability risks of negative interest rates while at the same time reaping in full the additional stimulating effects of the D-ZLB through the risk-appetite channel – this underlines that the additional amplification due to the D-ZLB is inherently linked to the increase in bank failure risk due to the D-ZLB, as shown in Section 2. Still, a monetary policy cut to -0.31% with a simultaneous increase in the capital requirement to 9% stimulates bank lending beyond the level of bank lending that the policy rate cut would achieve in absence of the D-ZLB (as well as the loan supply level that a cut to 0% would achieve, compare Table 8), while keeping bank failure risk relatively low. Although a normative analysis is beyond the scope of this paper, these results suggest that combining negative interest rate policy with an increase in capital requirements can be a reasonable policy. While the state-dependent changes in monetary policy and capital requirements are anticipated by banks here, the conclusions are robust to unexpected changes in monetary policy and capital requirements (see Online Appendix).

Table 5: *Increase of Capital Requirements in state P or N*

Policy Rate (%)	-0.31	-0.31	-0.31	-0.31	-0.31
Capital Requirement (%, State P)	8.0	9.0	10.0	8.0	8.0
Capital Requirement (%, State N)	8.0	8.0	8.0	9.0	9.5
Loan Volume (P)	52.483	49.546	46.69	52.596	52.645
Loan Volume (N)	46.458	46.718	46.896	42.849	41.187
Bankruptcy Prob. (%, P)	0.539	0.398	0.29	0.526	0.521
Bankruptcy Prob. (%, N)	0.766	0.749	0.738	0.551	0.485
Share Constrained (P)	0.195	0.197	0.199	0.191	0.189
Share Constrained (N)	0.38	0.373	0.367	0.379	0.38
Deposit Insurance Costs (P)	0.085	0.056	0.036	0.085	0.085
Deposit Insurance Costs (N)	0.109	0.11	0.11	0.07	0.058

Note: This table reports the average aggregate loan volume, bank failure probability, share of capital-constrained banks and deposit insurance cost by aggregate state $s_t \in \{P, N\}$ for different combinations of state-dependent capital requirements. The first column corresponds to the baseline.

4 Discussion

The simulations of the quantitative model presented above highlight two novel results: first, monetary policy is on average *more* expansionary in negative territory in the presence of a D-ZLB, at least initially. Second, NIRP entails substantial risks for financial stability, due to sizable increases in the bank-failure probability in the presence of a D-ZLB. Both results are two sides of the same coin, and arise because sufficiently capitalized banks charge lower loan rates on risky loans via a novel amplification channel of negative interest rates ("risk-appetite channel"), established in the stylized model of Section 2, that is operative when banks have some probability of failure. To the best of my knowledge, none of the theoretical contributions assessing NIRP (e.g. Abadi et al., 2023; Ulate, 2021; Eggertsson et al., 2024) has considered risky lending with imperfectly correlated loan defaults, precluding this channel from operating in these papers, and precluding them from assessing financial stability risks.[34]

Relation to the Empirical Literature. The risk-appetite channel identified in this paper helps rationalize several findings in the empirical literature on negative interest rate policy (NIRP).

First, it provides an explanation for the evidence that banks more exposed to negative

[34] As discussed previously in Footnote 3, the channel cannot arise with perfectly correlated loan defaults since in that case the losses in deposit-taking cannot impact the conditional-on-solvency return on lending.

rates increase lending relative to less exposed banks after policy rates fall below zero (e.g. Demiralp et al., 2021; Hong and Kandrac, 2021; Bottero et al., 2022; Schelling and Towbin, 2022). In the model, exposure to the deposit zero lower bound (D-ZLB) raises bank failure risk and thereby increases the marginal value of limited liability. Unconstrained banks respond by lowering loan rates and expanding credit supply, implying a stronger lending response.

Second, the model offers an explanation of the portfolio-rebalancing effects documented by Heider et al. (2019), Bottero et al. (2022), and Basten and Mariathasan (2023). Interpreting S as safe lending, a decline in loan rates under a binding D-ZLB increases the share of risky loans in banks' portfolios, since the aggregate deposit base $\bar{D}$ is fixed. The resulting shift toward riskier assets resembles the portfolio reallocation observed in the data. While Heider et al. (2019) attribute their findings primarily to a risk-taking channel operating through franchise values, as in Repullo (2004), the mechanism proposed here provides an alternative and potentially complementary explanation.

The empirical literature often characterizes these patterns as reflecting a search-for-yield or portfolio-rebalancing motive (Hong and Kandrac, 2021; Demiralp et al., 2021; Bottero et al., 2022). The model confirms this intuition while clarifying its underlying source. The increase in lending is not driven by banks' desire to reduce holdings of safe assets that yield negative returns. If that were the mechanism, the same effect would arise if banks had a zero default probability. Instead, lending expands because the D-ZLB raises bank failure risk, which increases the marginal limited-liability subsidy and induces banks to accept lower loan spreads. The resulting reallocation toward risky lending is therefore a consequence of greater risk-taking incentives generated by the interaction of negative rates and limited liability.

An important implication is that the model links the increase in lending by more exposed banks and the observed portfolio reallocation to a rise in bank riskiness. This is consistent with empirical evidence documenting higher bank risk of more exposed banks under negative policy rates (e.g. Hong and Kandrac, 2021). Indeed, in the model the increase in bank risk is not a by-product of greater lending; it is the mechanism that generates the lending response.

Arguably, a higher loan supply of more exposed banks cannot be explained by models of NIRP that abstract from bank risk, such as Ulate (2021), Abadi et al. (2023), and

Eggertsson et al. (2024). In those frameworks, explaining this result generally requires systematic differences in loan demand elasticities or balance-sheet composition across the two groups. Consider, for example, the model of Abadi et al. (2023). If banks with greater deposit funding also held larger portfolios of fixed-rate long-term assets, policy-rate cuts would increase the market value of those assets and temporarily strengthen their equity positions, allowing them to sustain higher lending.[35] However, the same mechanism would imply that these banks should also experience stronger equity gains following policy-rate cuts in positive territory.

This implication appears difficult to reconcile with the evidence. For instance, Hong and Kandrac (2021) report placebo exercises based on stock-price reactions to policy-rate cuts above zero (Appendix C). Since stock prices reflect changes in the market value of bank equity,[36] the balance-sheet-composition explanation would predict stronger reactions among highly exposed banks even outside negative-rate territory. Instead, the authors find abnormal stock-price responses for high-deposit banks when rates are cut into negative territory, while the corresponding effects following positive-rate cuts are substantially smaller and generally statistically insignificant.

Finally, the existence of the risk-appetite channel is not inconsistent with studies documenting declines in bank lending under negative rates (Basten and Mariathasan, 2023; Eggertsson et al., 2024; Heider et al., 2019). As discussed in the introduction, such findings may reflect either (i) portfolio reallocation away from the particular loan category examined in a given study—for example, syndicated lending in Heider et al. (2019) or household lending in Eggertsson et al. (2024)—or (ii) a sufficiently large share of banks being constrained by regulatory capital requirements.[37]

Policy Implications. Despite the on average higher loan supply under negative interest rate policy due to the D-ZLB in the quantitative results – reflecting the dominance of

[35] As emphasized by Abadi et al. (2023), equity erosion through deposit-taking losses is gradual, whereas bond revaluation effects are immediate.

[36] Changes in the number of shares and dividend payouts are unlikely to matter because Hong and Kandrac (2021) study a 40-minute window around policy announcements.

[37] Heider et al. (2019) report a suggestive robustness exercise using aggregate loan portfolios of euro-area banks, although data limitations preclude a formal regression analysis. By contrast, Demiralp et al. (2021), using data on banks' total loan portfolios, find that more exposed euro-area banks increased lending relative to their peers. Similar conclusions emerge from the credit-registry evidence of Bottero et al. (2022) for Italy and Grandi and Guille (2023) for France. Taken together, these findings suggest that a portfolio-reallocation interpretation of Heider et al. (2019)'s results is plausible and consistent with the mechanism proposed here.

the risk-appetite channel – these also suggest substantial heterogeneity between banks. Some banks are forced to restrict lending in the presence of a D-ZLB to fulfill the capital requirement due to losses from deposits eroding equity, as in Abadi et al. (2023). For this reason, negative interest rate policy may become less expansionary on average over time, as in Ulate (2021). In the simulations of the model calibrated for Germany presented above, this never happens for monetary policy rates of -0.31% (the baseline calibration) as long as rates are kept negative, while for policy rates of -0.5% it happens after 6 years. The simulations thus suggest that the ECB's spell of negative interest rates between 2014 and 2022 with a trough of -0.5% that was kept between 2019 and 2022 likely amplified lending (on average) for German banks beyond the effect that a conventional monetary policy cut would have had. While the simulations predict that over longer time horizons the equity erosion effect emphasized by e.g. Abadi et al. (2023) becomes stronger such that the additional stimulatory effect of negative interest rate policy fades off and loan volumes fall below the level that would be expected for a conventional monetary policy rate cut, the results suggest that the cuts into negative policy rates still remain expansionary, with a throughout higher aggregate loan supply under a negative policy rate after a cut to both -0.31% and to -0.5% even after 50 years of such negative policy rates.

On the other hand, my results suggest a large impact on the bank-failure probability of negative interest rate policy. Therefore, supervisors should be particularly vigilant about bank failure risk in times of NIRP, and should consider accompanying policy rate cuts in negative territory with an increase in capital requirements (see Section 3.7).

5 Conclusion

This paper has studied the effects of negative monetary policy rates on banks in the presence of a zero lower bound on deposit rates (D-ZLB). Banks' hesitance to set negative deposit rates has been a salient finding of empirical studies of the introduction of negative policy rates in various advanced economies since the mid 2010s. This paper highlights for the first time that this lack of transmission into deposit rates changes the loan supply decision of banks not constrained by capital requirements when banks are risky. This is because loan supply decisions are affected by a bank's failure probability, which in turn is

affected by deposit rates. I show in a simple model that such unconstrained banks charge lower loan rates on risky loans: an amplification effect of negative interest rates that works through a novel risk-appetite channel. However, this comes at a cost in terms of financial stability, with the bank-failure probability increasing due to the losses from deposits. It is important to note that this novel risk-appetite channel for *unconstrained* banks is not in contradiction with the reversal result of Abadi et al. (2023), which is due to banks being constrained by capital requirements. The novel mechanism can however explain empirical findings of banks more exposed to negative interest rates increasing their credit supply compared to less exposed banks (e.g. Hong and Kandrac, 2021; Demiralp et al., 2021; Bottero et al., 2022).

After establishing these result formally in a stylized model, the paper proceeds to quantify the relative importance of the various effects of negative monetary policy rates, namely the amplifying effect of the risk-appetite channel and the equity erosion effect highlighted by previous contributions (Ulate, 2021; Abadi et al., 2023), as well as their impact on financial stability, in a quantitative model of the banking industry. The model is designed to quantify the effects of a lower bound on deposits on loan supply and financial stability – noting that in the absence of such a D-ZLB, there is no difference between conventional and negative interest rate policy in the model.

In the quantitative model, calibrated for Germany, the amplifying effect of the risk-appetite channel is large and dominates the other effects, such that aggregate loan volumes are on average about 9% higher under negative interest rates in the presence of a D-ZLB. At the same time, negative interest rate policy implies a substantial deterioration of financial stability under a D-ZLB, with the bank-failure probability increasing by 30 basis points under negative policy rates of -0.31% – constituting a very sizable 64% increase.

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Online Appendix

A Quantitative Model: Robustness & Extensions

This section presents results from various variations of the model as robustness exercises. Each variation of the baseline model is independent from the others.

A.1 Higher Cost of Equity Injections

I first test the sensitivity of the results to the parameter governing the cost of bank equity injections χ. The parameter is critical for the importance of the equity-erosion channel: the larger χ, the more costly it is for banks to restore their equity and hence their lending capacity when their equity level is low.

To assess the sensitivity of the results to this, I set (i) $\chi = 2.5$ and (ii) $\chi = 5$. The remaining parameters are unchanged. The results for $\chi = 2.5$ are presented in Table 6 and those for $\chi = 5$ are presented in Table 7.

The equity-erosion channel is indeed stronger than in the baseline for both $\chi = 2.5$ and $\chi = 5$: while the unconstrained loan rate remains almost unchanged (it is 1 bp lower than in the baseline), aggregate loan volumes are somewhat lower when the cost of equity injections is larger, as expected. For $\chi = 2.5$ aggregate loan volumes are on average 8% higher due to the D-ZLB, somewhat smaller than in the baseline calibration (where it was 9%). For $\chi = 5$, this effect is reduced further to 7.6%. Hence, while the equity-erosion channel does get stronger as the cost of equity injection increases, the risk-appetite channel still dominates even for very large values of χ (about 7 times larger than in the baseline) for a policy rate of -0.31% in state N.

A.2 Higher Excess Cost of Bank Capital

I next test the sensitivity of the results to the choice of the excess cost of bank capital Δ_E, for which the available empirical evidence points to a relatively large time-series variance (European Central Bank, 2015). In the baseline calibration, I set $\Delta_E = 0.05$. Instead, I now set an 8% p.a. excess cost of bank capital, i.e. $\Delta_E = 0.08$. All other parameters are kept fixed. The results are reported in Table 15. Expectedly, the higher

excess cost of capital leads to larger loan rate spreads in both states. The size of the D-ZLB effect on the loan rates, however, is unchanged compared to the baseline calibration (11 bps in state N). As in the baseline calibration, aggregate loan supply is higher in the D-ZLB scenario in both states. However, the effect on the average bank-failure probability under negative policy rates is larger now compared to the baseline: the default probability in state N doubles from 0.37% to 0.75% due to the D-ZLB.

A.3 Varying Loan Default Probabilities

I next assess the role of varying loan default probabilities. In the baseline model, loan default probabilities are constant. However, loan default probabilities typically vary over the business cycle, and can be much higher in recessions (see e.g. Mendicino et al. 2026). To gauge the importance of this for the size of D-ZLB effects, I now introduce a second aggregate state of the economy $s_{2t} \in \{H, L\}$, governing times of high and low firm default probabilities $p(s_{2t})$. For simplicity, I assume that s_{2t} follows a Markov chain that is independent from the Markov chain s_t that governs policy rates and aggregate productivity. Hence, there are now four aggregate states of the economy: H, P (high firm default probability, and high productivity with a positive policy rate); H, N (high firm default probability, and low productivity with a negative policy rate); L, P (low firm default probability, and high productivity with a positive policy rate) and L, N (low firm default probability, and low productivity with a negative policy rate).

I set $p(H) = 0.0485$ and $p(L) = 0.0234$, corresponding to the average firm default probability in the euro area when the firm default probability is above and below the 90th percentile respectively, as reported in Mendicino et al. (2026). Additionally, Claessens et al. (2009) report an average duration of credit-crunch episodes of about 2.5 years. Hence, I set the probability of a transition from state H to state L to 0.4 to get an average duration of the high-firm-default state (in which banks extend especially low credit volumes) of 2.5 years. Given that I set the firm default probability in state H to its conditional average above the 90th percentile, I set the probability of transitioning from state L to state H to 0.0444 to match an average frequency of 10% for the high-firm-default state. All other parameters are kept fixed.

The results are reported in Table 16. The table suggests that the higher average loan default rate in state H significantly strengthens the impact of the D-ZLB on the

unconstrained loan rate when policy rates are negative (that is, in state H, N). In that case, the unconstrained loan rate is 3.1% in absence of a D-ZLB, compared to 2.52% when banks are subject to a D-ZLB. On the other hand, when the firm default probability is low in the negative policy rate state (state L, N) the unconstrained loan rate is 1.84% in absence of a D-ZLB and decreases to 1.78% due to the D-ZLB – the amplification due to the risk-appetite channel is thus much smaller when the firm default probability is low. This is intuitive: if the firm-default probability was zero, the risk-appetite channel would disappear entirely.

At the same time, the D-ZLB effect on the bank-failure probability is much stronger when the firm-default probability is high: the bank-failure probability increases by 66 bps in state H, N, whereas the increase is only 20 bps in state L, N. Nevertheless, aggregate loan supply is higher in the D-ZLB scenario in all states. Hence, this exercise illustrates that a change in the distribution of loan default rates (within the observed range of average loan default rates in the euro area) has a quantitatively important impact on the size of the D-ZLB effect.

A.4 Loan Default Probabilities as a Function of the Loan Rate

Up to now, loan default probabilities were assumed to be independent of loan default rates. In reality, loan rates affect loan default probabilities through at least three channels: (i) adverse selection (Stiglitz and Weiss, 1981), (ii) borrower moral hazard, with borrowers optimally increasing their own default risk when facing higher loan rates (Stiglitz and Weiss, 1981; Boyd and De Nicolo, 2005), and (iii) general equilibrium feedback effects through prices (Kiyotaki and Moore, 1997; Bernanke et al., 1999; Brunnermeier and Sannikov, 2014). Empirically, Uppal (2025) shows that monetary tightening increases loan delinquency of floating rate loans, thus providing evidence for loan default probabilities increasing in loan rates.

Banks internalize the impact of loan rates on the loan default probabilities, hence their optimal loan rates will be affected by such a feedback loop. To gauge the impact on the effects of the D-ZLB, I now assume that the individual borrowing firm's unconditional default probability is given by $p_t = max(a + b(R_{Lt} - 1), 1)$, following Martinez-Miera and Repullo (2010). The strength of the feedback between loan rates and the unconditional loan default probability is thus governed by b.

I set $a = 0.02$ and $b = 0.15$, such that the unconditional default probability is above (below) the baseline value of 0.027 in state P (N). The value of b implies that for a 1 ppa higher loan rate, the loan default rate rises by 15 bps, which is comparable to the evidence from Uppal (2025): he shows that a 1pp increase in the monetary policy rate is associated with a hump-shaped increase in loans past-due of (as a percentage of total assets) which ranges from approximately 5 to 20 basis point within the first year after the contraction and rises to a maximum of 50bps (but recall that the quantitative model is at yearly frequency).

The model is re-estimated to match the same targets as the baseline calibration. The calibration results are presented in Table 10. The resulting parameters are: $\alpha = 0.945$, $\rho = 0.263$, $A(P) = 0.196$, $A(N) = 0.154$, $\chi = 0.75$. The remaining parameters are unchanged.

The results are presented in Table 17. Additionally, the histogram of loan volume differences due to the D-ZLB is depicted in Figure 14.

In state N, i.e. under negative interest rates, the unconstrained loan rate is lower compared to the baseline in both scenarios (D-ZLB and no D-ZLB), reflecting the now decreased loan default probability in that state. The effect of the D-ZLB on the unconstrained loan rate is also lower than in the baseline (approximately 8 bps), and the average loan volume only increases by around 4% in state N.

The effect of the D-ZLB on the bank-failure probability is also weaker than in the baseline model. In state N, it increases by 20 bps due to the D-ZLB, up from a level of 0.36%.

Thus, as in the baseline model, the risk-appetite channel still dominates over the equity-erosion channel, and the D-ZLB increases bank riskiness. However, the feedback channel between loan rates and loan default rates weakens the effects significantly.

A.5 Lower Elasticity of Loan Demand

The elasticity of loan demand for a given loan rate is governed by α and δ. However, there is tension between matching loan spreads and the elasticity of lending, which is too high in the model compared to empirical estimates by Altavilla et al. (2022), who find a decrease in loan demand between 1 and 2% following a 100bps increase in their lending rate. Decreasing the elasticity of loan demand (by decreasing α or increasing δ) increases

the loan spread that a monopolist bank chooses (since when it marginally raises the loan rate, the resulting decrease in loan volumes is smaller, and hence the bank chooses to set a higher spread). I consider matching loan spreads to be more important, as the payoff of performing loans greatly impacts the implications of negative interest rate policy for bank default risk in the model and this is of first order importance for both the effect on loan supply of unconstrained banks and financial stability, as demonstrated in the illustrating model.

It may be argued that this impacts the relative importance of the loan supply effect of unconstrained and constrained banks, with my model overstating the importance of the effect of unconstrained banks. To evaluate this argument, I set $\alpha = 0.8$. I set D $= 0.54$ to maintain a comparable loan to deposit ratio to the baseline. I furthermore set $\rho = 0.37$, allowing me to match a bank default probability of 0.7%. The remaining parameters remain unchanged.

The main results are presented in Table 11. With these parameters, banks choose a loan rate around 4.45% in the negative policy rate state and of 7.9% in the positive state, implying a loan volume around 0.5 in both states and a loan-demand elasticity of about 4.5% in equilibrium in the negative interest rate state.

The D-ZLB effects on loan supply and bank defaults are qualitatively unchanged. In state N, the bank default probability is 15 bps higher due to the D-ZLB, compared to 30 bps in the baseline, and aggregate loan volumes are 3% higher, compared to 9% in the baseline. While these results are quantitatively weaker, this affects both the bank default probability and the loan volumes, indicating that most of the effect is not due to a weakening of the risk-appetite channel in response to the lower demand elasticity. Rather, these lower impacts reflect the much higher payoff from performing loans, which is protecting banks from defaults, and hence weakening the impact of the D-ZLB on risk-taking incentives compared to the baseline.

A.6 Low Deposit Banks

In the baseline calibration used for the main quantitative results above, banks have abundant deposits and choose to invest a part of them at the policy rate in order to exploit market power in loan markets. While this is appropriate for the average German bank (see Table 1), it is less appropriate for the average euro-area bank. As Table 14

shows, the average loan-to-deposit was well above 1 for euro-area banks during the spell of negative interest rate policy conducted by the ECB. In the context of the model here, such banks would thus borrow at the policy rate ($S_{kt} < 0$). However, in reality, banks with a failure probability above zero would not be able to borrow at the riskless rate $R(s_t)$. In fact, the interest rates at which commercial banks can borrow from the ECB were never negative, and accessing the facility requires posting collateral.[38] In unsecured transactions on the interbank market, on the other hand, borrowing banks with a non-zero failure probability would have to compensate lenders for default risk by paying a higher interest rate (see e.g. Afonso et al., 2011).

Therefore, I now assume that banks can borrow funds S^- on financial markets, such that the balance sheet of bank k is now: $L_{kt} + S_{kt} = D_{kt} + S^-_{kt} + E_{kt}$. The interest rate that lenders charge on non-deposit financing S^-_{kt} is denoted $\tilde{R}_{kt}$. I further assume that this debt is junior to deposits, such that the funds available to repay non-deposit lenders are given by:[39]

$$\Omega_{kt} = max\{[(1-\omega_k)R_{Lkt} + \omega_k(1-\lambda)]L_{kt} - R_{Dt}D_{kt}, 0\}.$$

Hence, the expected return for lenders that provide non-deposit debt to bank k is:

$$R^S_{kt} = \tilde{R}_{kt} - \frac{\int_0^1 max\{\tilde{R}_{kt}S^-_{kt} - \Omega_{kt}, 0\}dF(\omega)}{S^-_{kt}}. \tag{A.1}$$

I assume that lenders are risk-neutral, and may alternatively invest in the safe asset at rate R. Lenders are thus willing to lend S^-_{kt} to bank k if they break even in expectation:

$$R^S_{kt} = R_t. \tag{A.2}$$

In this subsection I consider an alternative calibration to approximately match the average loan-to-deposit ratios of euro-area banks. To this end, the deposit base is reduced to

[38] The ECB sets two key interest rates at which banks can borrow from the ECB against collateral: the main refinancing operation and marginal lending facility.

[39] This is in line with EU regulation. Article 108 of the EU Bank Recovery and Resolution Directive 2014/59/EU grants eligible deposits of natural persons and micro, small and medium-sized enterprises priority over claims of unsecured creditors. Eligible deposits, as defined in Article 2, No (95) of the directive, are all deposits not listed in Article 5 of the Directive 2014/49/EU, which provides exclusions for deposits associated with e.g. money laundering. Thus, for the purposes of this model, all deposits are assumed eligible.

$\bar{D} = 40$, $A(P)$ is set to 0.1971 and $A(N)$ is set to 0.156, ρ is set to 0.258 and $\chi = 0.25$. The remaining parameters are unchanged. The corresponding model and data moments are summarized in Table 14.

The results are summarized in Table 18. A histogram of differences in bank lending between the baseline scenario in which banks are subject to a D-ZLB and the counterfactual in which they are not is depicted in Figure 15. The amplifying effect of the risk-appetite channel is of a similar magnitude as in the baseline model, although somewhat stronger: the unconstrained loan rate is 12 bps lower in the presence of a D-ZLB in the negative monetary policy state. Once again, the risk-appetite channel dominates over the equity-erosion channel in the quantitative results, with the aggregate loan volume being on average 13.5% higher in the presence of a D-ZLB in state N. The effect on the bank-failure probability is somewhat weaker than in the baseline, with an increase of 25 basis points due to the D-ZLB under negative policy rates (corresponding to a 50% increase).

A.7 Deposit Insurance

So far, it has been assumed that deposits are fully insured. This appears a good approximation for Germany (for which the main analysis is calibrated), where apart from the legal minimum guarantee of 100,000 Euro per depositor a large proportion of banks are covered by additional institutional deposit protection (Deutsche Bundesbank, 2015), as well as ample implicit government guarantees – for example, there was a political guarantee for 100% of private savings deposits during the financial crisis in 2008 (ifo Institute, 2011).

To gauge how important the assumption of full deposit insurance is, I repeat the analysis assuming that (a) 20% and (b) 40% of deposits are uninsured. Let D_t^I and D_t^U denote insured and uninsured deposits, respectively, with corresponding deposit rates R_{Dt}^I, R_{Dt}^U. In line with reality, uninsured deposits are assumed to be junior to insured deposits (see Footnote 39). Hence, available funds to repay uninsured depositors are (assuming $S > 0$, which is the case in the baseline calibration):

$$\Sigma_{kt} = max\{[(1-\omega_k)R_{Lkt} + \omega_k(1-\lambda)]L_{kt} + R_t S_t - R_{Dt}^I D_{kt}^I, 0\},$$

so that the expected return for lenders that provide uninsured deposits D^U_{kt} to bank k is:

$$\tilde{R}^U_{Dt} = R^U_{Dt} - \frac{\int_0^1 min\{\tilde{R}^U_{Dt} D^U_{kt} - \Sigma_{kt}, 0\} dF(\omega)}{D^U_{kt}}. \tag{A.3}$$

I assume that uninsured depositors are risk-neutral, and demand the same return as insured depositors:

$$\tilde{R}^U_{Dt} = R^I_{Dt} \tag{A.4}$$

All parameters are identical to the baseline model.

The results are presented in Table 19 (for a share of 20% uninsured deposits) and Table 20 (for a share of 40% uninsured deposits). The conclusions are qualitatively unchanged and quantitatively similar. The average deposit rate across insured and uninsured deposits is only slightly higher (approximately 2 bps) compared to the baseline scenario without uninsured deposits. This is true for both a share of 20% and 40% of uninsured deposit funding: since a higher share of uninsured deposit funding implies that the bank has more funds available after paying the (senior) insured deposit debt, the rate on uninsured deposits is actually lower when the share of these deposits is 40% than when it is 20%. As a consequence of the limited impact on the average deposit rate, the impact on the D-ZLB effect on bank default risk and loan supply is also small: in state N (negative interest rate policy) the effect on bank riskiness is 34 bps (the baseline result was 30 bps) and average loan volumes are 10% higher under the D-ZLB (the baseline result was 9%).

A.8 Unexpected Monetary Policy & Macroprudential Policy

This section presents additional results in which the monetary policy change from state P to N is unexpected. This has the advantage that the policy function of banks (i.e. their loan rates, dividends and equity injections as a function of their equity) are identical in state P both in the scenario in which banks cannot lower deposit rates into negative territory as well as in the counterfactual without D-ZLB. The results are presented in Table 12. Additionally, various combinations of macroprudential and monetary policy that have been considered in the main text are presented in Table 13. These exercises confirm the conclusions of Section 3.4 of the main text.

B Additional Tables and Figures

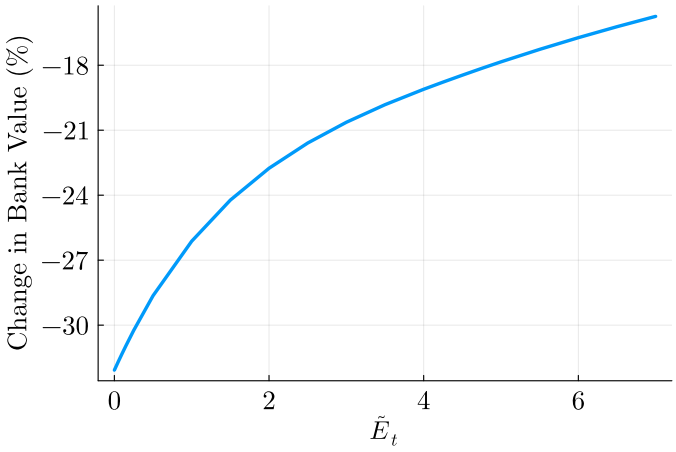
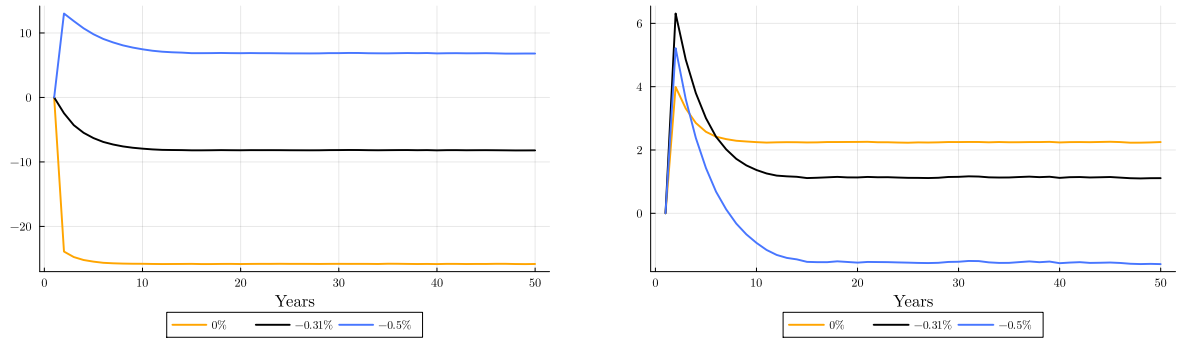


Figure 8: *Price Impact and Equity*

The figure shows the percent change in the bank's value function $V(s, \tilde{E})$ when the exogenous state of the economy s changes from P to N: $100 \times \left[V(N, \tilde{E}) - V(P, \tilde{E})\right] / V(P, \tilde{E})$.



(a) *Mean Loan IRF* **(b)** *Difference Loans IRFs (D-ZLB - no D-ZLB)*

Figure 9: *Evolution After Permanent Productivity Change From $A(P)$ To $A(N)$*

Note: This figure depicts the average evolution across islands k of selected variables after a permanent change from state P to state N for varying policy rates in state N. The policy rate in state P, as well as the sequence of loan default rates ω_{kt} is kept constant across simulations.

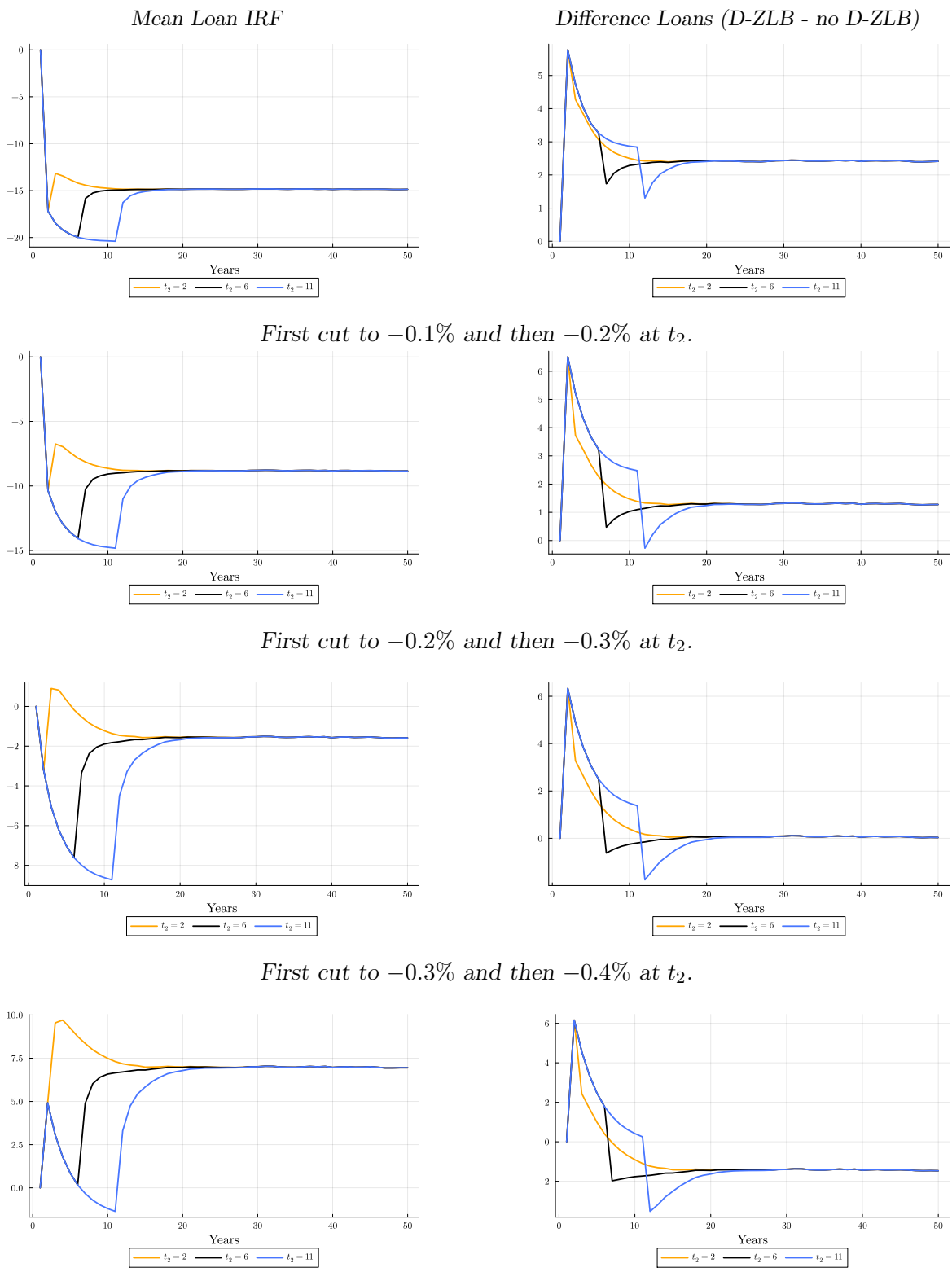


Figure 13: *IRFs to Marginal Cuts*

The figure shows the evolution after a productivity change from $A(P)$ to $A(N)$ with a simultaneous first monetary policy cut at $t = 1$ and a second cut at t_2. The left panel depicts the IRF of lending starting from state P, while the right panel shows the differences in the IRF of the baseline D-ZLB economy and the counterfactual no D-ZLB economy.

Table 6: *Average Results By State ($\chi = 2.5$)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	52.02	45.8	51.51	42.36
Loan Rate (unconst., %)	4.92	1.93	4.94	2.05
Avg. Loan Rate (const. %)	5.41	2.34	5.42	2.45
Avg. Loan Rate (%)	5.03	2.1	5.03	2.12
Share Constrained (%)	20.67	39.92	17.59	15.63
Bank Failure Prob. (%)	0.54	0.77	0.51	0.47
Deposit Insurance Costs	0.08	0.11	0.08	0.06
Deposit Insurance Costs (Unconditional)		0.091		0.075
Loan Volume (Unconditional)		50.136		48.74
Loan Volume Difference (State N - P)		-6.21		-9.15
D-ZLB Bank Risk Effect	0.02	0.31		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 7: *Average Results By State ($\chi = 5$)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	51.82	45.54	51.44	42.31
Loan Rate (unconst., %)	4.92	1.93	4.94	2.05
Avg. Loan Rate (const. %)	5.44	2.38	5.45	2.47
Avg. Loan Rate (%)	5.04	2.12	5.03	2.12
Share Constrained (%)	21.11	40.62	17.78	15.79
Bank Failure Prob. (%)	0.53	0.79	0.51	0.46
Deposit Insurance Costs	0.08	0.11	0.08	0.06
Deposit Insurance Costs (Unconditional)		0.091		0.075
Loan Volume (Unconditional)		49.915		48.671
Loan Volume Difference (State N - P)		-6.27		-9.12
D-ZLB Bank Risk Effect	0.02	0.32		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 8: *Average Results By State ($R(N) = 0$)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	0.0	2.37	0.0
Deposit Rate (%)	1.35	0.0	1.35	-0.99
Loan Volume	51.47	27.66	51.04	24.15
Loan Rate (unconst., %)	4.93	2.33	4.94	2.45
Avg. Loan Rate (const. %)	5.33	2.68	5.34	2.82
Avg. Loan Rate (%)	5.02	2.4	5.03	2.48
Share Constrained (%)	24.48	20.73	22.93	7.15
Bank Failure Prob. (%)	0.52	0.54	0.5	0.3
Deposit Insurance Costs	0.08	0.04	0.08	0.02
Deposit Insurance Costs (Unconditional)		0.071		0.063
Loan Volume (Unconditional)		44.254		42.888
Loan Volume Difference (State N - P)		-23.82		-26.89
D-ZLB Bank Risk Effect	0.02	0.24		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 9: *Average Results By State ($R(N) = -0.005$)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.5	2.37	-0.5
Deposit Rate (%)	1.35	0.0	1.35	-1.49
Loan Volume	52.96	62.12	51.76	59.71
Loan Rate (unconst., %)	4.92	1.73	4.95	1.82
Avg. Loan Rate (const. %)	5.34	2.06	5.37	2.14
Avg. Loan Rate (%)	5.0	1.88	5.01	1.89
Share Constrained (%)	17.95	45.96	15.62	21.54
Bank Failure Prob. (%)	0.54	0.78	0.52	0.51
Deposit Insurance Costs	0.09	0.15	0.08	0.09
Deposit Insurance Costs (Unconditional)		0.105		0.086
Loan Volume (Unconditional)		55.735		54.167
Loan Volume Difference (State N - P)		9.17		7.95
D-ZLB Bank Risk Effect	0.02	0.28		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 10: *Moments & Targets (Endogenous Loan Default Probability Model of Section A.4)*

Moment	Data	Model
Mean Loan Spread	2.56	2.39
Mean Bank Default	0.66	0.55
L/D (State P)	1.03	1.03
L/D (State N)	0.92	0.98
Equity Injections to Total Equity	0.3	0.33

Note: The loan spread is calculated as the difference of corporate lending rates as reported by the ECB (2003-2023, excluding 2008/2009) and the relevant policy rate; bank-failure probability from Mendicino et al. (2026) for the euro area; Deposit-to-Loan Ratio for the euro area from ECB BIS (1999Q1-2023Q3); Equity injections are calculated from National Accounts data from Eurostat, see Footnote 30.

Table 11: *Results with $\alpha = 0.8$*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	0.53	0.502	0.524	0.487
Loan Rate (unconst., %)	7.9	4.45	7.94	4.52
Avg. Loan Rate (const. %)	8.22	4.79	8.25	4.99
Avg. Loan Rate (%)	7.9	4.44	7.94	4.52
Share Constrained (%)	1.71	1.08	1.49	1.29
Bank Failure Prob. (%)	0.7	0.72	0.63	0.57
Loan Volume (Unconditional)		0.521		0.513
Loan Volume Difference (State N - P)		-0.028		-0.037
D-ZLB Bank Risk Effect	0.07	0.15		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 12: *Unexpected Monetary Policy Cuts*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	51.63	46.33	52.14	42.78
Loan Rate (unconst., %)	4.94	1.94	4.94	2.05
Avg. Loan Rate (const. %)	5.36	2.27	5.35	2.39
Avg. Loan Rate (%)	5.02	2.07	5.01	2.1
Share Constrained (%)	19.12	38.03	17.01	15.26
Bank Failure Prob. (%)	0.53	0.76	0.52	0.47
Deposit Insurance Costs	0.08	0.11	0.08	0.06
Deposit Insurance Costs (Unconditional)		0.091		0.077
Loan Volume (Unconditional)		50.02		49.30
Loan Volume Difference (State N - P)		-5.3		-9.36
D-ZLB Bank Risk Effect	0.02	0.3		

Note: The table reports average results from a model in which the change from P to N is unexpected (MIT shock). Results are computed over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 13: *Unexpected Monetary Policy Cuts & Macroprudential Policy*

Policy Rate (%)	-0.31	-0.31	-0.31	-0.31	0.0
D-ZLB	No	Yes	Yes	Yes	Yes
Capital Requirement (%, State P)	8.0	8.0	8.0	8.0	8.0
Capital Requirement (%, State N)	8.0	8.0	8.5	9.0	7.0
Loan Volume (P)	52.136	51.631	51.678	51.721	50.871
Loan Volume (N)	42.781	46.329	44.473	42.745	26.545
Bankruptcy Prob. (%, P)	0.516	0.534	0.528	0.522	0.508
Bankruptcy Prob. (%, N)	0.469	0.764	0.647	0.551	0.452
Deposit Insurance Costs (P)	0.084	0.083	0.083	0.083	0.081
Deposit Insurance Costs (N)	0.06	0.109	0.087	0.07	0.035

Note: This table reports the average aggregate loan volume, bank failure probability, share of capital-constrained banks and deposit insurance cost by aggregate state $s_t \in \{P, N\}$ for different combinations of state-dependent capital requirements when the change from P to N is unexpected (MIT shock). The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 14: *Moments & Targets (Low Deposit Model of Section A.6)*

Moment	Data	Model
Mean Loan Spread	2.62	2.45
Mean Bank Default	0.66	0.69
L/D (State P)	1.31	1.36
L/D (State N)	1.12	1.09
Equity Injections to Total Equity	1.03	0.93

Note: The loan spread is calculated as the difference of corporate lending rates as reported by the ECB (2003-2023, excluding 2008/2009) and the relevant policy rate; bank-failure probability from Mendicino et al. (2026) for the euro area; Deposit-to-Loan Ratio for the euro area from ECB BIS (1999Q1-2023Q3); Equity injections are calculated from National Accounts data from Eurostat, see Footnote 30.

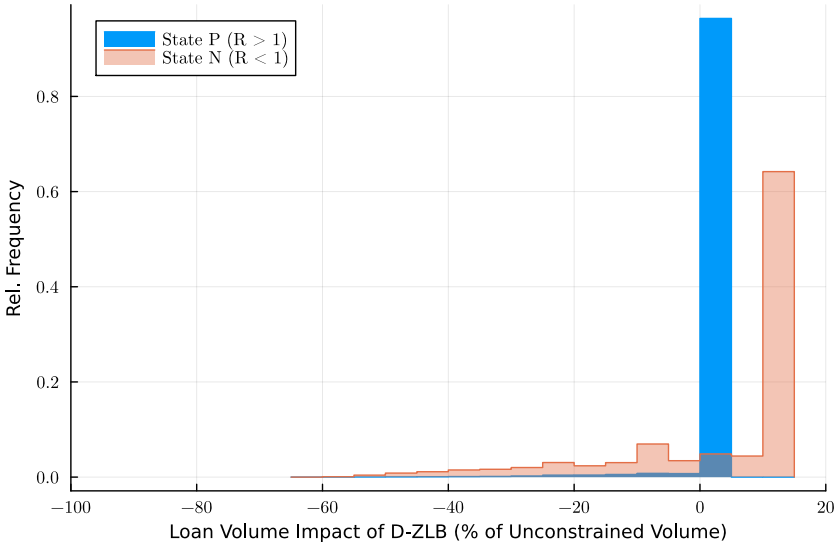


Figure 14: *Histogram Loan Supply Differences when Loan Default Probabilities are a Function of the Loan Rate (Section A.4))*

This figure depicts the distribution of differences $L_{kt}^{DZLB} - L_{kt}^{noDZLB}$ in loan supply in the scenario in which banks are subject to a D-ZLB and the scenario in which they are not subject to a D-ZLB, by aggregate state. The sequence of loan default rates ω_{kt} on a given island k is kept constant across simulations.

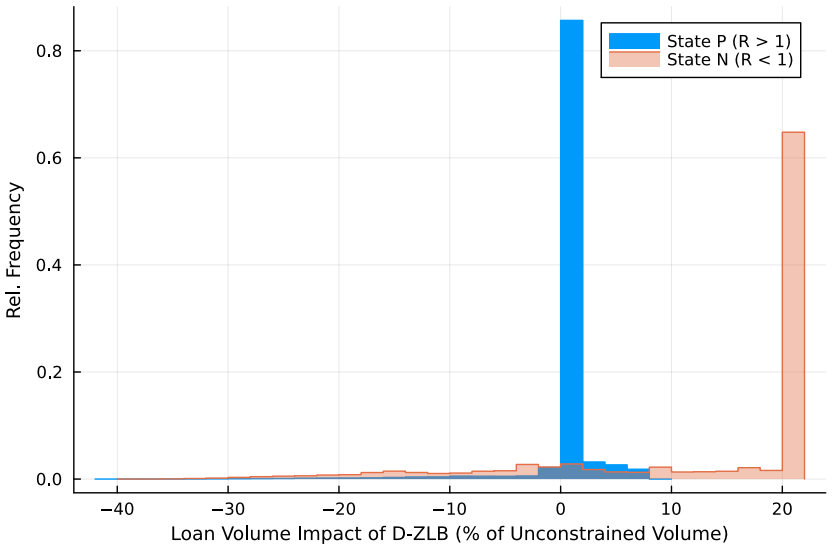


Figure 15: *Histogram Loan Supply Differences (Low Deposit Model of Section A.6)*

This figure depicts the distribution of differences $L_{kt}^{DZLB} - L_{kt}^{noDZLB}$ in loan supply in the scenario in which banks are subject to a D-ZLB and the scenario in which they are not subject to a D-ZLB, by aggregate state. The sequence of loan default rates ω_{kt} on a given island k is kept constant across simulations.

Table 15: *Average Results By State (8% Excess Cost of Bank Capital)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.35	0.0	1.35	-1.3
Loan Volume	38.97	32.32	38.6	29.37
Loan Rate (unconst., %)	5.18	2.2	5.2	2.31
Avg. Loan Rate (const. %)	5.59	2.5	5.61	2.65
Avg. Loan Rate (%)	5.25	2.3	5.25	2.35
Share Constrained (%)	15.52	33.64	12.62	10.03
Bank Failure Prob. (%)	0.47	0.75	0.44	0.37
Deposit Insurance Costs	0.05	0.07	0.05	0.03
Deposit Insurance Costs (Unconditional)		0.06		0.047
Loan Volume (Unconditional)		36.955		35.806
Loan Volume Difference (State N - P)		-6.65		-9.23
D-ZLB Bank Risk Effect	0.03	0.37		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 16: *Average Results By State (Varying Firm Default Probability p, Section A.3)*

	D-ZLB				No D-ZLB			
Firm Default Prob. (%)	2.34	4.85	2.34	4.85	2.34	4.85	2.34	4.85
Policy Rate (%)	2.37	2.37	-0.31	-0.31	2.37	2.37	-0.31	-0.31
Deposit Rate (%)	1.36	1.36	0.0	0.0	1.36	1.36	-1.3	-1.3
Loan Volume	64.38	15.87	57.55	20.66	63.54	14.27	56.34	9.78
Loan Rate (unconst., %)	4.72	5.98	1.78	2.52	4.74	6.08	1.84	3.1
Avg. Loan Rate (const. %)	5.11	6.69	2.09	3.06	5.13	6.79	2.15	3.66
Avg. Loan Rate (%)	4.85	6.0	1.92	2.6	4.86	6.09	1.92	3.11
Share Constrained (%)	29.26	2.99	46.02	16.04	28.29	2.14	26.23	1.77
Bank Failure Prob. (%)	0.42	0.4	0.57	0.91	0.4	0.33	0.37	0.25
Deposit Insurance Costs	0.08	0.02	0.1	0.06	0.08	0.01	0.06	0.0
Deposit Insurance Costs (Unconditional)		0.081		0.067				
Loan Volume (Unconditional)		58.514		57.583				
D-ZLB Bank Risk Effect	0.02	0.07	0.2	0.66				

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 17: *Average Results By State when Loan Default Probabilities are a Function of the Loan Rate (Section A.4)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.36	0.0	1.36	-1.3
Loan Volume	51.4	48.85	50.55	46.82
Loan Rate (unconst., %)	4.87	1.67	4.89	1.75
Loan Rate (const. %)	5.29	1.96	5.31	2.05
Loan Rate (%)	4.94	1.78	4.95	1.79
Share Constrained (%)	17.95	38.57	15.87	14.14
Bankruptcy Prob. (%)	0.55	0.56	0.53	0.36
Deposit Insurance Costs	0.09	0.08	0.09	0.05
Deposit Insurance Costs (Unconditional)		0.08		0.07
Loan Volume (Unconditional)		50.45		49.17
ZLB Bank Risk Effect	0.02	0.2		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 18: *Average Results By State (Low Deposit Model of Section A.6)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.36	0.0	1.36	-1.3
Loan Volume	54.48	43.72	54.52	38.53
Loan Rate (unconst.)	4.96	1.91	4.96	2.03
Loan Rate (const.)	5.38	2.16	5.39	2.26
Loan Rate	4.97	1.99	4.97	2.06
Share Constrained	14.96	35.41	14.21	14.4
Bankruptcy Prob. (Percent)	0.62	0.77	0.59	0.52
Deposit Insurance Costs	0.1103	0.1093	0.1097	0.0649
Deposit Insurance Costs (Unconditional)		0.1098		0.0877
Loan Volume (Unconditional)		49.19		46.66
ZLB Bank Risk Effect	0.02	0.25		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 19: *Average Results By State (20% Uninsured Deposits, Section A.7)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.37	0.02	1.37	-1.29
Uninsured Deposit Rate	1.45	0.11	1.45	-1.24
Loan Volume	51.77	46.41	51.16	42.29
Loan Rate (unconst.)	4.94	1.94	4.96	2.07
Loan Rate (const.)	5.32	2.24	5.33	2.34
Loan Rate	5.01	2.05	5.02	2.1
Share Constrained	18.44	39.65	15.17	11.88
Bankruptcy Prob. (Percent)	0.35	0.54	0.31	0.2
Deposit Insurance Costs	0.0505	0.075	0.0487	0.0234
Deposit Insurance Costs (Unconditional)		0.0625		0.0363
Loan Volume (Unconditional)		49.14		46.8
ZLB Bank Risk Effect	0.03	0.34		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

Table 20: *Average Results By State (40% Uninsured Deposits, Section A.7)*

	D-ZLB		No D-ZLB	
Policy Rate (%)	2.37	-0.31	2.37	-0.31
Deposit Rate (%)	1.37	0.02	1.37	-1.29
Uninsured Deposit Rate	1.4	0.06	1.4	-1.27
Loan Volume	51.77	46.4	51.16	42.28
Loan Rate (unconst.)	4.94	1.94	4.96	2.07
Loan Rate (const.)	5.32	2.24	5.33	2.34
Loan Rate	5.01	2.05	5.02	2.1
Share Constrained	18.44	39.65	15.15	11.89
Bankruptcy Prob. (Percent)	0.34	0.54	0.31	0.2
Deposit Insurance Costs	0.0497	0.075	0.0485	0.0235
Deposit Insurance Costs (Unconditional)		0.0621		0.0362
Loan Volume (Unconditional)		49.13		46.8
ZLB Bank Risk Effect	0.03	0.34		

Note: The table reports averages over a simulation of 1000 years and 10000 islands with independent shocks z_{kt}. Loan rate (const.) and Loan rate (unconst.) are the loan rates charged by, respectively, banks that restrict lending due to low equity levels, and those that have sufficient equity. Share Constrained is the share of banks that restrict lending due to low equity levels. The D-ZLB Bank Risk Effect is the change in the bankruptcy probability due to the deposit ZLB.

C Proofs

Proof of Lemma 1. The objective function of the bank was given by:

$$\max_{R_L, R_D \geq \underline{R}_D} \int_0^1 \max[(R_L - R)L(R_L) + (R - R_D)D(R_D) - \omega R_L L(R_L), 0] dF(\omega). \quad \text{(C.1)}$$

To replace the max(·) function within the interval, note that the bank will fail (and thus receive zero) if $\omega > \tilde{\omega}$, where

$$\tilde{\omega}(R_L, \hat{R}_D(R, \underline{R}_D), R, \underline{R}_D) = \frac{(R_L - R)L(R_L) + (R - \hat{R}_D)\bar{D}}{R_L L(R_L)}.$$

Given the support of $\omega \in [0,1]$, the upper bound of integration is:

$$\bar{\omega} = \begin{cases} 0 \text{ if } \tilde{\omega} < 0, \\ \tilde{\omega} \text{ if } 0 \leq \tilde{\omega} \leq 1, \\ 1 \text{ if } \tilde{\omega} > 1. \end{cases} \tag{C.2}$$

The objective of the bank as a function of R_D for any given R_L can then be written as:

$$\mathcal{L}(R_D) = \int_0^{\bar{\omega}} [(R_L(1-\omega) - R)L(R_L) - (R_D - R)D(R_D)]dF(\omega). \tag{C.3}$$

If $\tilde{\omega} \in [0,1]$, the integrand evaluated at the upper bound is zero by definition of $\bar{\omega}$. If, on the other hand, $\tilde{\omega} > 1$ or $\tilde{\omega} < 0$, the upper bound is unaffected by marginal changes in the deposit rate. Thus, it follows from Leibniz Rule that:

$$\mathcal{L}'(R_D) = F(\bar{\omega}) \left[-D(R_D) - (R_D - R)D'(R_D)\right], \tag{C.4}$$

where $D'(R_D) = (-\epsilon_D)\frac{D(R_D)}{R_D}$ with $\epsilon_D < -1$, such that:

$$\mathcal{L}'(R_D) = F(\bar{\omega})(-\epsilon_D)\frac{D(R_D)}{R_D}\left[R - \frac{\epsilon_D - 1}{\epsilon_D}R_D\right]. \tag{C.5}$$

We have $\frac{(-\epsilon_D)D(R_D)}{R_D} > 0$ for any $R_D > 0$. Further, if the bank does not always fail we have $F(\bar{\omega}) > 0$, in which case:

$$\mathcal{L}'(R_D^*) \stackrel{!}{=} 0 \iff R_D^* = \frac{\epsilon_D}{\epsilon_D - 1}R. \tag{C.6}$$

Then any $R_D > R_D^*$ would feature $\mathcal{L}' < 0$ since $\epsilon_D < -1$. Conversely, any $R_D < R_D^*$ would feature $\mathcal{L}' > 0$, implying that the bank could reach higher profits by decreasing or increasing, respectively, R_D. So R_D^* is the unique maximizer of (C.3). Now, if $R_D^* \geq \underline{R}_D$, then R_D^* is also the banks' constrained optimal deposit rate in the presence of a lower bound $\underline{R}_D$ on the deposit rate. However, if $R_D^* < \underline{R}_D$, then the bank maximizes (B.1) at the corner $\underline{R}_D$, since $\mathcal{L}' < 0$ for any $R_D > R_D^*$, such that setting a deposit rate higher than $\underline{R}_D$ leads to lower profits, and the bank cannot set a deposit rate lower than $\underline{R}_D$.

Thus in general:

$$\hat{R}_D = \max(R_D^*, \underline{R}_D). \tag{C.7}$$

Further, $R_D^*(R)$ is a differentiable and increasing function with:

$$\frac{\partial R_D^*(R)}{\partial R} = \frac{\epsilon_D}{\epsilon_D - 1} > 0 \tag{C.8}$$

and for any given $\underline{R}_D$, there exists $R^* = \frac{\epsilon_D - 1}{\epsilon_D}\underline{R}_D$ s.t. $R_D^*(R) > \underline{R}_D$ for all $R > R^*$ and $R_D^*(R) < \underline{R}_D$ for all $R < R^*$. ■

Proof of Proposition 1. First consider the possible corner solutions.

I will first show that it is never optimal for the bank to set loan rates lower than R. By definition:

$$\Pi(R) \geq \Pi(R_L) \text{ for all } R_L < R,$$

since for all such R_L, profits from lending, given by

$$[(1-\omega)R_L - R)]L(R_L),$$

are negative for all ω. If the bank does not make losses in deposit taking, i.e. if

$$(R - \hat{R}_D)D(\hat{R}_D) \geq 0,$$

the inequality is strict: $\Pi(R) > \Pi(R_L)$. Otherwise, the bank fails with certainty for all $R_L < R$ and is indifferent between all such loan rates.

Now consider the other corner, i.e. an infinite loan rate. When setting an infinite loan rate (and thus extending zero loans), loan demand implies

$$\lim_{R_L \to \infty} R_L L(R_L) = \lim_{R_L \to \infty} R_L^{-\epsilon_L + 1} = 0,$$

since $\epsilon_L > 1$. But then clearly

$$\lim_{R_L \to \infty} \Pi(R_L) = \max[(R - \hat{R}_D)D(\hat{R}_D), 0].$$

Now focus on the case in which the bank does not make losses in deposit taking, i.e.

$$(R - \hat{R}_D)D(\hat{R}_D) \geq 0.$$

To shorten notation, define

$$\pi(\omega; R_L) = [R_L(1-\omega) - R]L(R_L) + (R - \hat{R}_D)D(\hat{R}_D),$$

such that $\Pi(R_L) = \mathbb{E}\pi(\omega; R_L)$. Since $\max(x, 0)$ is convex, Jensen's inequality implies that

$$\mathbb{E}\max[\pi(\omega; R_L), 0] \geq \max[\mathbb{E}\pi(\omega; R_L), 0].$$

But then by definition:

$$\mathbb{E}max(\pi(\omega; R_L), 0) \geq max(L(R_L)\int_0^1 (R_L(1-\omega) - R)dF(\omega) + \underbrace{(R - \hat{R}_D)D(\hat{R}_D)}_{\geq 0}, 0).$$

Hence,

$$\mathbb{E}max(\pi(\omega; R_L), 0) = \Pi(R_L) > (R - \hat{R}_D)D(\hat{R}_D) \text{ for all } R_L > \frac{R}{1 - \mathbb{E}\omega} \text{ s.t. } L(R_L) > 0.$$

Since $\mathbb{E}\omega \neq 1$, the set $[\frac{R}{1-\mathbb{E}\omega}, \infty)$ is not empty. Further, since for now we only consider the case in which banks do not face losses in deposit taking:

$$\lim_{R_L \to \infty} \Pi(R_L) = max((R - \hat{R}_D)D(\hat{R}_D), 0) = (R - \hat{R}_D)D(\hat{R}_D).$$

Since

$$\Pi(R) = \int_0^1 max\{-\omega L(R) + (R - \hat{R}_D)D(\hat{R}_D), 0\}dF(\omega),$$

it is easy to see that:

$$\lim_{R_L \to \infty} \Pi(R_L) > \Pi(R).$$

This guarantees that there must be a loan rate $R_L \in [\frac{R}{1-\mathbb{E}\omega}, \infty)$ that gives a higher payoff to the bank than setting an infinite loan rate and hence extending zero loans, and also a higher payoff than setting $R_L \leq R$.

Next, consider the case in which the bank faces losses in deposit taking, i.e.

$$(R - \hat{R}_D)D(\hat{R}_D) < 0.$$

Obviously, if the bank never breaks even for any loan rate, i.e. if:

$$(R_L - R)L(R_L) + (R - \hat{R}_D)D(\hat{R}_D) \leq 0 \text{ for all } R_L,$$

then $\Pi(R_L) = 0$ for all R_L. If that is not the case, i.e. if there exists a loan rate R_L such that:

$$(R_L - R)L(R_L) + (R - \hat{R}_D)D(\hat{R}_D) > 0,$$

then $\max_{R_L} \Pi(R_L) > 0$ since ω has full support on $[0, 1]$ and the bank enjoys limited liability. Furthermore, since we focus on the case $(R - \hat{R}_D)D(\hat{R}_D) < 0$, we have that:

$$\lim_{R_L \to \infty} \Pi(R_L) = max((R - \hat{R}_D)D(\hat{R}_D), 0) = 0.$$

Hence, unless the bank fails with certainty for any loan rate, there is a loan rate $R_L \in \mathbb{R}$ that gives a strictly positive payoff, and hence there must be an interior maximum.

Having established the conditions for existence of an interior solution, let us now turn to necessary conditions. Since $\Pi(R_L)$ is a differentiable function, it must be that at the interior maximum $\Pi'(R_L) = 0$. It was assumed that loan demand is given by

$$L(R_L) = AR_L^{-\epsilon_L}, \ \epsilon_L > 1,$$

such that

$$\frac{R_L L'(R_L)}{L(R_L)} = -\epsilon_L. \tag{C.9}$$

The FOC of the maximization problem is:

$$\Pi'(R_L) = \int_0^{\tilde{\omega}} [((1 - \omega)R_L - R)L'(R_L) + (1 - \omega)L(R_L)]dF(\omega) \stackrel{!}{=} 0. \tag{C.10}$$

Now define

$$\omega^{LL} = \int_0^{\tilde{\omega}} \frac{\omega}{F(\tilde{\omega})} dF(\omega),$$

such that the FOC becomes:

$$F(\tilde{\omega})[((1-\omega^{LL})R_L - R)L'(R_L) + (1-\omega^{LL})L(R_L)] = 0. \tag{C.11}$$

Factoring out $L'(R_L)$ and using (C.9) allows to write (C.10) as:

$$F(\tilde{\omega})L'(R_L)[((1-\omega^{LL})\left(\frac{\epsilon_L - 1}{\epsilon_L}\right)R_L - R] = 0. \tag{C.12}$$

The optimal loan rate $\hat{R}_L$ thus fulfills:

$$(1-\omega^{LL})\hat{R}_L = \frac{\epsilon_L}{\epsilon_L - 1}R > R. \tag{C.13}$$

■

Proof of Proposition 2.

The optimal loan rate $\hat{R}_L$ is characterized by Eq. (C.12). The equation defines $\hat{R}_L$ as an implicit function of R and $\underline{R}_D$:

$$\Xi(\hat{R}_L(R,\underline{R}_D), \hat{R}_D(R,\underline{R}_D), R, \underline{R}_D) = (1-\omega^{LL})\left(\frac{\epsilon_L - 1}{\epsilon_L}\right)\hat{R}_L - R = 0 \tag{C.14}$$

Let Ξ_i denote the derivative of Ξ with respect to its i'th argument. Differentiating Ξ with respect to $\underline{R}_D$ yields:

$$\Xi_1\frac{\partial \hat{R}_L}{\partial \underline{R}_D} + \Xi_2\frac{\partial \hat{R}_D}{\partial \underline{R}_D} + \Xi_4 = 0. \tag{C.15}$$

The derivatives of Ξ are as follows. First, we have:

$$\Xi_1 = \frac{\epsilon_L - 1}{\epsilon_L}\left[(1-\omega^{LL}) - \hat{R}_L\frac{\partial \omega^{LL}}{\partial R_L}\right] > 0. \tag{C.16}$$

The sign follows from the fact that $\hat{R}_L$ is the optimal loan rate. Ξ_1 must then be positive – if it was negative at $\hat{R}_L$, then $\hat{R}_L$ would be a local minimum. Further, it was assumed that parameters and functional forms are such that at the (unique) local maximum $\Xi_1 \neq 0$ to ensure that the implicit function theorem can be applied. Section D establishes some sufficient conditions for this assumption to hold.

Next, we have that

$$\Xi_2 = -\left(\frac{\epsilon_L - 1}{\epsilon_L}\right)\frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{\partial \bar{\omega}}{\partial R_D}R_L, \tag{C.17}$$

where from Eq. (6):

$$\frac{\partial \bar{\omega}}{\partial R_D} = -\frac{\bar{D}}{\hat{R}_L L(\hat{R}_L)}, \tag{C.18}$$

which follows from the restriction to cases involving $\bar{\omega} \in (0,1)$ in the statement of the proposition; and further:

$$\frac{\partial \hat{R}_D}{\partial R} = \begin{cases} 0 \text{ if } R < R^*, \\ \frac{\epsilon_D}{\epsilon_D - 1} \text{ if } R > R^* \\ \text{and else does not exist} \end{cases} \tag{C.19}$$

as shown in Lemma 1.

Lastly, we have that $\Xi_4 = 0$. Using these terms in (C.15) and rearranging then yields:

$$\frac{\partial \hat{R}_L}{\partial \underline{R}_D} = -\frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{\partial R_D}{\partial \underline{R}_D}\frac{\bar{D}}{L(\hat{R}_L)}\frac{1}{(1-\omega^{LL}) - \hat{R}_L\frac{\partial \omega^{LL}}{\partial R_L}} \begin{cases} < 0 \text{ if } R < R^*, \\ \text{ does not exist if } R = R^*, \\ = 0 \text{ else.} \end{cases} \tag{C.20}$$

■

Proof of Corollary 1. From Proposition 2 it is straightforward to show that a bank is more likely to fail under the D-ZLB:

$$\frac{\partial \bar{\omega}}{\partial \underline{R}_D} = \underbrace{\frac{\partial \bar{\omega}}{\partial R_D}}_{<0}\frac{\partial R_D}{\partial \underline{R}_D} + \underbrace{\frac{\partial \bar{\omega}}{\partial R_L}}_{>0}\underbrace{\frac{\partial R_L}{\partial \underline{R}_D}}_{\leq 0} \begin{cases} < 0 \text{ if } R < R^*, \\ \text{ does not exist if } R = R^*, \\ = 0 \text{ else.} \end{cases} \tag{C.21}$$

This follows from Eq. (C.18), Eq. (D.5) and Proposition 2. ■

Proof of Corollary 2. Differentiating Eq. (C.14) with respect to R yields:

$$\Xi_1\frac{\partial R_L}{\partial R} + \Xi_2\frac{\partial \hat{R}_D}{\partial R} + \Xi_3 = 0. \tag{C.22}$$

Ξ_1 and Ξ_2 have already been computed above in the proof of Proposition 2. Lastly, we

have that

$$\Xi_3 = -\frac{\epsilon_L - 1}{\epsilon_L}\frac{\partial \omega^{LL}}{\partial R}\hat{R}_L - 1, \tag{C.23}$$

where

$$\frac{\partial \omega^{LL}}{\partial R} = \frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{S}{\hat{R}_L L(\hat{R}_L)}. \tag{C.24}$$

Thus, the pass-through of the monetary policy rate into loan rates is given by:

$$\frac{\partial \hat{R}_L}{\partial R} = \frac{\frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{\epsilon_L - 1}{\epsilon_L}\left[\frac{S}{L(\hat{R}_L)} - \frac{\bar{D}}{L(\hat{R}_L)}\frac{\partial \hat{R}_D}{\partial R}\right] + 1}{\frac{\epsilon_L - 1}{\epsilon_L}\left[(1-\omega^{LL}) - \hat{R}_L\frac{\partial \omega^{LL}}{\partial R_L}\right]}. \tag{C.25}$$

Using the balance sheet identity $L + S = D$, (C.25) can be written as:

$$\frac{\partial \hat{R}_L}{\partial R} = \frac{\frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{\epsilon_L - 1}{\epsilon_L}\frac{\bar{D}}{L(\hat{R}_L)}\left(1 - \frac{\partial \hat{R}_D}{\partial R}\right) + 1 - \frac{\partial \omega^{LL}}{\partial \bar{\omega}}\frac{\epsilon_L - 1}{\epsilon_L}}{\frac{\epsilon_L - 1}{\epsilon_L}\left[(1-\omega^{LL}) - \hat{R}_L\frac{\partial \omega^{LL}}{\partial R_L}\right]}, \tag{C.26}$$

where by Eq. (C.19) and $\epsilon_D < -1$ we have that $1 - \frac{\partial \hat{R}_D}{\partial R} > 0$. For the derivative $\frac{\partial \omega^{LL}}{\partial \bar{\omega}}$ note that by definition: $\omega^{LL} = \int_0^{\bar{\omega}} \omega \frac{f(\omega)}{F(\bar{\omega})} d\omega$. Hence, by the Leibniz rule:

$$\frac{\partial \omega^{LL}}{\partial \bar{\omega}} = \bar{\omega}\frac{f(\bar{\omega})}{F(\bar{\omega})} - \int_0^{\bar{\omega}} \omega \frac{f(\omega)f(\bar{\omega})}{F(\bar{\omega})^2} d\omega. \tag{C.27}$$

Using the definition of ω^{LL}, this simplifies to:

$$\frac{\partial \omega^{LL}}{\partial \bar{\omega}} = (\bar{\omega} - \omega^{LL})\frac{f(\bar{\omega})}{F(\bar{\omega})} > 0. \tag{C.28}$$

Hence if $\frac{\partial \omega^{LL}}{\partial \bar{\omega}} < \frac{\epsilon_L}{\epsilon_L - 1}$ (where $\frac{\epsilon_L}{\epsilon_L - 1} > 1$), then $\frac{\partial \hat{R}_L}{\partial R} > 0$. This is a relatively weak sufficient condition, which is for example fulfilled by the uniform distribution $\omega \sim Unif[0,1]$, in which case $\frac{\partial \omega^{LL}}{\partial \bar{\omega}} = \frac{1}{2}$. ■

Proof of Corollary 3. Eq. (9) characterizes the loan rate as an (implicit) function $R_L(R, \bar{\omega})$ of the policy rate R and the bank failure cut-off $\bar{\omega}$. The bank will therefore

charge the same loan rate for a given policy rate and failure cutoff, independent of the deposit lower bound. This is because (as shown in Proposition 2) the effect of the deposit lower bound on the loan rate operates entirely through the change in the bank failure cutoff. A change in the deposit lower bound therefore does not affect the loan rate if the bank receives an additional payoff $\Delta = \bar{D}\left(\underline{R}_{D1} - \frac{\epsilon_D}{\epsilon_D - 1}R\right)$ from the government that exactly offsets the effect of the change in the deposit lower bound, such that:

$$\bar{\omega}(\underline{R}_{D1}, \Delta) = 1 - \frac{(\underline{R}_{D1} - R)\bar{D} + RL(R_L) - \Delta}{R_L L(R_L)} = 1 - \frac{(\frac{\epsilon_D}{\epsilon_D - 1} - 1)R\bar{D} + RL(R_L)}{R_L L(R_L)} = \bar{\omega}(\underline{R}_{D2}, 0),$$

and hence $\hat{R}_L = R_L(R, \bar{\omega}(\underline{R}_{D1}, \Delta)) = R_L(R, \bar{\omega}(\underline{R}_{D2}, 0)$.
Further, Lemma 1 implies that

$$\frac{\partial R_D}{\partial R} \begin{cases} = \frac{\epsilon_D}{\epsilon_D - 1} \text{ if } R > R^*, \\ \text{does not exist if } R = R^*, \\ = 0 \text{ if } R < R^*. \end{cases} \tag{C.29}$$

Using this in (C.26) – which was derived in the proof of Corollary 2 – it follows directly that:

$$\begin{aligned} &\left.\frac{\partial \hat{R}_L}{\partial R}\right|_{R^*(\underline{R}_{D1}) > R, R, \bar{\omega}} - \left.\frac{\partial \hat{R}_L}{\partial R}\right|_{R^*(\underline{R}_{D2}) < R, R, \bar{\omega}} \\ &= \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{\bar{D}}{L(\hat{R}_L)\left[(1 - \omega^{LL}) - \hat{R}_L \frac{\partial \omega^{LL}}{\partial R_L}|_{\hat{R}_L}\right]} \frac{\epsilon_D}{\epsilon_D - 1} > 0. \end{aligned} \tag{C.30}$$

■

D Uniqueness and Conditions of Implicit Function Theorem

In order for Proposition 2 to characterize transmission of monetary policy, it is required that the FOC Eq. (9) – denoted Ξ as in the previous section – admits a unique solution and the conditions of the implicit function theorem are fulfilled.

Applicability of the Implicit Function Theorem In particular, this requires (a) $\Xi(\hat{R}_L(R, \underline{R}_D), \hat{R}_D(R, \underline{R}_D), R, \underline{R}_D) = 0$, (b) its derivatives Ξ_1, Ξ_2, Ξ_3 and Ξ_4 to be continuous in an open set that contains the point $(R_L(R, \underline{R}_D), \hat{R}_D(R, \underline{R}_D), R, \underline{R}_D)$ and (c) $\Xi_1 \neq 0$. Condition (a) is fulfilled by definition for $\hat{R}_L(R, \underline{R}_D)$. The conditions for the existence of such a loan rate that solves the FOC have been established in Proposition 1. The relevant partial derivatives for Condition (b) were detailed in the proof of Proposition 2. Condition (b) is fulfilled as long as $\frac{\partial \omega^{LL}}{\partial \bar{\omega}}$ is continuous. This is because sums and products of continuous functions are continuous.

In the following, I shall therefore focus on condition (c), i.e. $\Xi_1 \neq 0$. While I do not show that the condition $\Xi_1 \neq 0$ holds in general, I establish it for a specific example. Concretely, it shall be shown that if $\frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2} \geq 0$ and $R < \underline{R}_D$, no local maximum can involve $\Xi_1 \leq 0$. To see this, first note that from Eq. (C.12):

$$\Pi''(\hat{R}_L) = L'(R_L) F(\bar{\omega}) \Xi_1.$$

Since $L'(R_L) < 0$ and for any bank that does not fail with certainty $F(\bar{\omega}) > 0$ (which was assumed throughout), it follows directly that no interior local maximum can involve $\Xi_1 < 0$, since this would imply $\Pi''(\hat{R}_L) > 0$, a contradiction. Next, assume that $\Pi''(\hat{R}_L) = 0$. But then it can be shown that $\Pi'''(\hat{R}_L) \neq 0$, such that $\hat{R}_L$ is a saddle point not a local maximum. To see this, first use the definition of Ξ_1 to obtain:

$$\Pi''(\hat{R}_L) = L'(R_L) F(\bar{\omega}) \frac{\epsilon_L - 1}{\epsilon_L} \left[(1 - \omega^{LL}) - R_L \frac{\partial \omega^{LL}}{\partial R_L} \right]. \tag{D.1}$$

Then, the third derivative of the profit function evaluates to:

$$\begin{aligned} \Pi'''(\hat{R}_L) = &\left[L''(R_L) F(\bar{\omega}) + L'(R_L) f(\bar{\omega}) \frac{\partial \bar{\omega}}{\partial R_L} \right] \underbrace{\left[(1 - \omega^{LL}) - R_L \frac{\partial \omega^{LL}}{\partial R_L} \right]}_{=0 \text{ by assumption}} + \\ &+ L'(R_L) F(\bar{\omega}) \underbrace{\frac{\epsilon_L - 1}{\epsilon_L} \left[-2 \frac{\partial \omega^{LL}}{\partial R_L} - \frac{\partial^2 \omega^{LL}}{\partial R_L^2} R_L \right]}_{=\Xi_{11} \text{ by (C.16)}}. \end{aligned} \tag{D.2}$$

As long as the bank does not always fail, $L'(R_L) F(\bar{\omega}) < 0$, hence we can focus on the

sign of Ξ_{11}. By definition, we know that:

$$\frac{\partial \omega^{LL}}{\partial R_L} = \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{\partial \bar{\omega}}{\partial R_L}, \tag{D.3}$$

and:

$$\frac{\partial \bar{\omega}}{\partial R_L} = \frac{[(1-\bar{\omega})R_L - R]L'(R_L) + (1-\bar{\omega})L(R_L)}{R_L L(R_L)}. \tag{D.4}$$

Using the assumption of iso-elastic loan demand, Eq. (D.4) can be written as:

$$\frac{\partial \bar{\omega}}{\partial R_L} = \frac{1}{R_L}\left[(1-\bar{\omega})(1-\epsilon_L) + \frac{R}{R_L}\epsilon_L\right]. \tag{D.5}$$

Note that at the optimum:

$$\frac{R}{R_L} = (1-\omega^{LL})\frac{\epsilon_L - 1}{\epsilon_L},$$

such that:

$$\left.\frac{\partial \omega^{LL}}{\partial R_L}\right|_{R_L=\hat{R}_L} = \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{\left[(1-\bar{\omega})(1-\epsilon_L) + (1-\omega^{LL})(\epsilon_L - 1)\right]}{R_L} \tag{D.6}$$

$$= \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{(\omega^{LL} - \bar{\omega})(1-\epsilon_L)}{R_L} \geq 0, \tag{D.7}$$

where the sign follows from $\epsilon_L > 1$, Eq. (C.28) and the fact that $\omega^{LL} \leq \bar{\omega}$, with strict inequality whenever $\bar{\omega} \in (0,1)$.

Next, we have:

$$\frac{\partial^2 \omega^{LL}}{\partial R_L^2} = \frac{\partial \omega^{LL}}{\partial \bar{\omega}} \frac{\partial^2 \bar{\omega}}{\partial R_L^2} + \frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2}\left(\frac{\partial \bar{\omega}}{\partial R_L}\right)^2, \tag{D.8}$$

where from Eq. (D.5):

$$\frac{\partial^2 \bar{\omega}}{\partial R_L^2} = -\frac{\partial \bar{\omega}}{\partial R_L}\left[\frac{1}{R_L}(1-\epsilon_L)\right] - \left[\frac{(1-\bar{\omega})(1-\epsilon_L)}{R_L^2} + 2\frac{R}{R_L^3}\epsilon_L\right] \tag{D.9}$$

$$\implies \frac{\partial^2 \bar{\omega}}{\partial R_L^2} R_L = -\frac{\partial \bar{\omega}}{\partial R_L}(1-\epsilon_L) - \frac{\partial \bar{\omega}}{\partial R_L} - \frac{R}{R_L^2}\epsilon_L, \tag{D.10}$$

where the second line once again uses Eq. (D.5). Using Eq. (D.8) and (D.10), it follows

that:

$$-2\frac{\partial\omega^{LL}}{\partial R_L}-\frac{\partial^2\omega^{LL}}{\partial R_L^2}R_L=-2\frac{\partial\omega^{LL}}{\partial R_L}+\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\left[\frac{\partial\bar{\omega}}{\partial R_L}(1-\epsilon_L)+\frac{\partial\bar{\omega}}{\partial R_L}+\frac{R}{R_L^2}\epsilon_L\right] \\ -R_L\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\left(\frac{\partial\bar{\omega}}{\partial R_L}\right)^2. \tag{D.11}$$

Now using Eq. (D.3) in (D.11) allows to simplify this to:

$$=\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\left[\frac{\partial\bar{\omega}}{\partial R_L}(-\epsilon_L)+\frac{R}{R_L^2}\epsilon_L\right]-R_L\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\left(\frac{\partial\bar{\omega}}{\partial R_L}\right)^2, \tag{D.12}$$

where ϵ_L can be factored out of the first term:

$$=(-\epsilon_L)\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\left[\frac{\partial\bar{\omega}}{\partial R_L}-\frac{R}{R_L^2}\right]-R_L\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\left(\frac{\partial\bar{\omega}}{\partial R_L}\right)^2. \tag{D.13}$$

Using Eq. (D.5), this can be expressed as:

$$-2\frac{\partial\omega^{LL}}{\partial R_L}-\frac{\partial^2\omega^{LL}}{\partial R_L^2}R_L=(-\epsilon_L)(1-\epsilon_L)\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\left[\frac{(1-\bar{\omega})}{R_L}-\frac{R}{R_L^2}\right]-R_L\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\left(\frac{\partial\bar{\omega}}{\partial R_L}\right)^2. \tag{D.14}$$

By the definition of $\bar{\omega}$, this is equivalent to:

$$-2\frac{\partial\omega^{LL}}{\partial R_L}-\frac{\partial^2\omega^{LL}}{\partial R_L^2}R_L=\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\frac{(R_D-R)D(R_D)}{R_L^2L(R_L)}\epsilon_L(\epsilon_L-1)-R_L\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\left(\frac{\partial\bar{\omega}}{\partial R_L}\right)^2. \tag{D.15}$$

Using Eq. (D.6) allows to write this as:

$$-2\frac{\partial\omega^{LL}}{\partial R_L}-\frac{\partial^2\omega^{LL}}{\partial R_L^2}R_L=\frac{\partial\omega^{LL}}{\partial\bar{\omega}}\epsilon_L(\epsilon_L-1)\frac{(R_D-R)D(R_D)}{R_L^2L(R_L)}-\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\frac{(\omega^{LL}-\bar{\omega})^2(\epsilon_L-1)^2}{R_L}. \tag{D.16}$$

Plugging this back into Eq. (D.2) shows that at any loan rate that fulfills the FOC:

$$\Pi'''(R_L)=L'(R_L)F(\bar{\omega})\left[\frac{\partial\omega^{LL}}{\partial\bar{\omega}}(\epsilon_L-1)^2\frac{(R_D-R)D(R_D)}{R_L^2L(R_L)}+\frac{\partial^2\omega^{LL}}{\partial\bar{\omega}^2}\frac{(\omega^{LL}-\bar{\omega})^2}{R_L}\frac{(\epsilon_L-1)^3}{\epsilon_L}\right]. \tag{D.17}$$

Now, first consider the case of a uniform distribution ($\frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2} = 0$). In this case, clearly:

$$\Pi'''(\hat{R}_L) = 0 \iff R = \underline{R}_D.$$

Recall that by definition of $\hat{R}_L$, $\Pi'(\hat{R}_L) = 0$. Hence, $R \neq \underline{R}_D$ is a sufficient (but not necessary) condition for $\Pi'''(\hat{R}_L) \neq 0$ under a uniform distribution of ω, implying that any such point that also involves $\Pi''(\hat{R}_L) = 0$ is a point of inflection and cannot be an optimum. Since no local maximum can involve $\Pi''(\hat{R}_L) > 0$ either, it must be that $\Pi''(\hat{R}_L) < 0$ at any local maximum $\hat{R}_L$ whenever $R \neq \underline{R}_D$.

More broadly, note that under negative interest rates ($R < \underline{R}_D$), the first term in the square brackets of (D.17) is positive. The second term is positive or zero if $\frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2} \geq 0$ (since $\epsilon_L > 1$). Hence, if $\frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2} \geq 0$ and $\bar{\omega} \in (0,1)$: $\Pi''(\hat{R}_L) = 0 \implies \Pi'''(\hat{R}_L) < 0$. This contradicts that such a $\hat{R}_L$ is a local maximum. Hence, under policy rates $R < \underline{R}_D$ any local maximum must involve $\Pi''(\hat{R}_L) < 0$ if $\frac{\partial^2 \omega^{LL}}{\partial \bar{\omega}^2} \geq 0$ (and cannot involve $\Pi''(\hat{R}_L) = 0$).

Uniqueness: Example Assume the conditions of Proposition 1 are fulfilled, such that some R_L exists that solves the FOC:

$$(1 - \omega^{LL}) \left(\frac{\epsilon_L - 1}{\epsilon_L} R_L \right) - R = 0.$$

For the sake of providing an example, consider $\omega \sim Unif[0,1]$. Focus on the case of a bank with an interior bank-failure probability $\bar{\omega} \in (0,1)$, since this is the case which Proposition 2 focuses on. Using the fact that under a uniform distribution $\omega^{LL} = \frac{1}{2}\bar{\omega}$ for any $\bar{\omega} \in (0,1)$, the FOC becomes:

$$\left(1 - \frac{1}{2}\bar{\omega}\right) \left(\frac{\epsilon_L - 1}{\epsilon_L} R_L \right) = R. \tag{D.18}$$

By definition of $\bar{\omega}$:

$$\left(1 - \frac{1}{2}\bar{\omega}\right) R_L = R_L - \frac{1}{2} R_L \left[1 - \frac{(R_D - R)D(R_D) + RL(R_L)}{R_L L(R_L)} \right] \tag{D.19}$$

$$= \frac{1}{2} \left[R_L + (R_D - R) \frac{D(R_D)}{L(R_L)} + R \right] \tag{D.20}$$

$$= \frac{1}{2} \left[R_L + (R_D - R) D(R_D) A R_L^{\epsilon} + R \right], \tag{D.21}$$

where the last line uses the definition of $L(R_L)$. Clearly, this is increasing in R_L if $R_D - R \geq 0$ (i.e. when $R \leq \underline{R}_D$, such that the bank faces losses in deposit taking.). A sufficient condition for a unique maximum under a uniform distribution is therefore $R \leq \underline{R}_D$.

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